



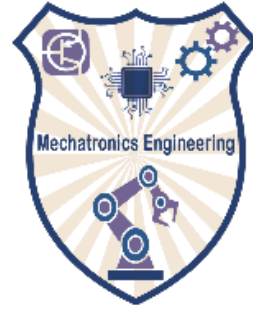
University of Jordan
School of Engineering

ROBOTIC SYSTEMS



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الجامعة الأردنية – كلية الهندسة

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**Analytical Forward and Inverse Kinematics
of 6-Dof-F Robotic Manipulators**

(0918474)

Robotic Systems

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Table of Responsibilities:

Table 1: Responsibilities and tasks assigned to each team member

Student Name	Role in the Project	Detailed Contributions
Rama Haddad	Kinematics Algorithm Development & Analytical Modeling	- Developed the complete forward kinematics (FK) and inverse kinematics (IK) models for all three robotic manipulators using an analytical approach. - Carefully derived the Denavit–Hartenberg (DH) parameters for each manipulator, ensuring correct frame assignment, joint orientation, and coordinate consistency. - Implemented the full transformation matrix formulation and verified correctness through systematic matrix multiplication and frame validation. - Solved the inverse kinematics analytically, extracting all joint variables step-by-step while handling multiple solutions, singular configurations, and joint constraints. - Performed symbolic simplification of the kinematic equations to improve clarity, computational efficiency, and numerical stability. - Ensured forward–inverse consistency, validating that FK results fed into IK return the original joint configurations accurately. - Structured and documented the kinematics code in a clean, modular format suitable for integration with the GUI and visualization modules. - Acted as the theoretical backbone of the project by translating robotic kinematics theory into a fully functional and verified implementation.
Meera Shersheer	GUI Development, System Integration & Visualization	- Designed and implemented the Graphical User Interface (GUI) that allows users to interactively control and visualize the robotic manipulators. - Integrated the kinematics algorithms with the GUI, ensuring seamless data flow between user inputs, computation modules, and visual outputs. - Connected the 2D and 3D visual representations of the manipulators to the mathematical model, allowing real-time updates of position and orientation. - Developed user-friendly controls for joint variables, end-effector positioning, and manipulator selection. - Ensured accurate synchronization between the analytical kinematics, CAD models, and graphical visualization. - Optimized the GUI layout for clarity, usability, and professional presentation suitable for academic demonstration.

		• Played a key role in system-level integration, transforming separate theoretical and CAD components into a single cohesive application. • Diagnosed and resolved complex integration issues arising from mismatched coordinate frames, data formats, and update timing between kinematics code, CAD models, and the GUI environment. • Served as the primary troubleshooting and integration lead, ensuring that all subsystems operated reliably together under different user inputs and test scenarios.
Own Al-Mazahreh	CAD Modeling & Mechanical Representation	• Designed detailed 2D and 3D CAD models for all three robotic manipulators, maintaining mechanical realism and accurate proportions. • Defined all link dimensions, joint types, axes of rotation, and reference frames in alignment with the kinematic model. • Ensured precise correspondence between CAD geometry and DH parameters, allowing correct physical interpretation of the kinematic equations. • Created clean and well-structured CAD assemblies suitable for both visualization and educational analysis. • Paid special attention to frame placement, joint alignment, and coordinate consistency, which is critical for correct forward and inverse kinematics validation. • Exported and prepared CAD models for seamless integration into the GUI and visualization environment. • Provided the mechanical foundation of the project, ensuring that the mathematical model represents a realistic and physically valid robotic system.

Although each team member had specific responsibilities as described above, the final report and PowerPoint presentation were completed together, with all members contributing to reviewing, improving, and finalizing the project.

I. Abstract

Robotic manipulators with five and six degrees of freedom are widely used in industrial and research applications due to their flexibility and high dexterity. Accurate kinematic modeling, particularly inverse kinematics, is essential for precise positioning, motion control, and trajectory planning of such manipulators. However, deriving analytical inverse kinematics solutions become increasingly complex as the number of degrees of freedom increases.

In this project, analytical forward and inverse kinematics equations are derived for three different robotic manipulators with six degrees of freedom. The Denavit–Hartenberg convention is employed to model the manipulator geometry and derive the homogeneous transformation matrices. Closed-form inverse kinematics solutions are obtained using geometric and algebraic approaches, while considering multiple solution branches, joint limits, and singular configurations.

The derived kinematic models are implemented using Python, and their correctness is validated by comparing inverse kinematics results with forward kinematics outputs for multiple test configurations. A unified graphical user interface is developed to integrate all forward and inverse kinematics computations, allowing user-defined inputs and real-time visualization of the manipulators in both two-dimensional and three-dimensional spaces.

The results demonstrate accurate agreement between forward and inverse kinematics solutions, confirming the validity of the derived analytical models. The developed framework provides a reliable and user-friendly platform for kinematic analysis of multi-degree-of-freedom robotic manipulators and can be extended to more complex robotic systems.

Keywords: *Inverse kinematics, Forward kinematics, Robotic manipulators, Denavit–Hartenberg parameters, Analytical modeling*

II. Table of Contents

I. Abstract	3
II. Table of Contents.....	4
III. List of Figures	6
IV. List of Tables.....	7
V. List of Abbreviations.....	8
VI. Introduction	9
A. Problem Background	9
B. Existing Solution Methods and Limitations	9
C. Motivation and Contributions of This Work	9
D. Selected Industrial Manipulators	10
Manipulator 1: UR5 – Universal Robots	10
Manipulator 2: ABB IRB 1600.....	13
Manipulator 3: KUKA KR16.....	15
VII. Problem Statement and Formulation	18
A. Problem Definition.....	18
B. Mathematical Formulation.....	18
C. Assumptions.....	24
D. Algorithm Description	25
VIII. Solution Methodology.....	27
A. Parameters, Variables, and Coefficients	27
B. Algorithm Implementation.....	27
C. Software Implementation and Graphical User Interface	27
C.1 Input Interface	27
C.2 Output Interface.....	28
C.3 CAD Visualization Module.....	31
IX. Results and Discussion	33
A. Numerical Examples and Results	33
Manipulator 1: UR5 – Universal Robots	33
Manipulator 2: ABB IRB 1600.....	40
Manipulator 3: (e.g., KUKA KR16)	43
B. Comparison of Solution Methods	48
i. Tables	48
C. Analysis and Discussion of Results	48
i. Interpretation:	48
ii. Strengths and Weaknesses:	48
iii. Observations:.....	48
D. Numerical Forward Kinematics Validation	48
X. Conclusions and Future Work	50

A. Summary of Findings.....	50
B. Limitations	50
C. Future Research Directions.....	50
XI. References.....	51
XII.Appendices	52
Appendix A: Forward Kinematics Code.....	52
Appendix B: Inverse Kinematics Code:	52

III. List of Figures

Figure 1: UR5 3D CAD Model.....	10
Figure 2: UR5 3D and 2D drawings including its parameters.....	11
Figure 3: DH parameters showed on the manipulator UR5 itself showing its joint frames	12
Figure 4: ABB IRB 1600 3D CAD Model	13
Figure 5 : ABB IRB 1600 3D and 2D drawings including its parameters	14
Figure 6: DH parameters showed on the manipulator ABB IRB 1600 itself showing its joint frames.....	15
Figure 7: KUKA KR16 3D CAD Model.....	16
Figure 8: KUKA KR 16 3D and 2D drawings including its parameters	16
Figure 9: DH parameters showed on the manipulator KUKA KR16 itself showing its joint frames.....	17
Figure 10 : 3D CAD Model for the three selected manipulators	18
Figure 11 Flow chart for the Robotic Manipulator Kinematics Analysis Platform.....	25
Figure 12: Graphical user interface showing the input tab for manipulator selection.....	28
Figure 13 Output interface showing symbolic forward kinematics transformation for UR5	29
Figure 14 Robotic Manipulator Kinematics Analysis Platform – Input Interface.....	30
Figure 15 Symbolic Inverse Kinematics Target End-Effector Pose Representation.....	30
Figure 16 Inverse Kinematics Solution Steps: Wrist Center and Base Joint Computation.....	30
Figure 17 Inverse Kinematics Joint Angle Derivation and Configuration Analysis	31
Figure 18 : 3D CAD model visualization tab showing the three-dimensional representation	32
Figure 19: 2D Sections tab showing different 2d views , projections and some manufacturer details	32
Figure 20: Entering case 1 inputs on the designed Platform	35
Figure 21: The GUI's results for case 1 showing the input joint values and individual link transformation..	35
Figure 22: The GUI's results for case 1 showing the output over all transformation and the end-effector pose	36
Figure 23: The GUI's results for case 2 showing the input and a part of the output.....	39
Figure 24: The GUI's results for case 2 showing the multiple solutions as the output with the error percentage	39
Figure 25: The GUI's results for case 3 showing the output over all transformation and the end-effector pose	40
Figure 26: The GUI's results for case 4 showing the multiple solutions as the output with the error percentage	42
Figure 27 Overall forward kinematics transformation matrix and resulting end-effector position and orientation.	43
Figure 28 Forward kinematics results showing input joint values and individual link transformation matrices.	44
Figure 29 Inverse kinematics result indicating an unreachable target pose.....	47
Figure 30: Numerical forward kinematics results in rad for the UR5 manipulator	49
Figure 31: Numerical forward kinematics results in deg for the UR5 manipulator	50

IV. List of Tables

Table 1: Responsibilities and tasks assigned to each team member 1

Table 2 : DH Table for the manipulator UR5 12

Table 3: DH Table for ABB IRB 1600 manipulator..... 14

Table 4: DH Table for KUKA KR16 manipulator 17

Table 6. Kinematic parameters and variables used in the analysis.....27

Table 7. Comparison between desired end-effector pose and FK validation48

Table 8. Comparison between analytical and numerical inverse kinematics48

V. List of Abbreviations

Abbreviation	Description
FK	Forward Kinematics
IK	Inverse Kinematics
DOF	Degree of Freedom
DH	Denavit–Hartenberg
EE	End Effector
GUI	Graphical User Interface
CAD	Computer-Aided Design
RPY	Roll–Pitch–Yaw
HTM	Homogeneous Transformation Matrix
(T)	Homogeneous Transformation Matrix
(T_0^n)	Transformation Matrix from Base Frame to Frame n
(R)	Rotation Matrix
(p)	Position Vector
(x, y, z)	Cartesian Position Coordinates
(α, β, γ)	Roll, Pitch, and Yaw Angles
(θ_i)	Joint Angle of Joint i
(a_i)	Link Length
(d_i)	Link Offset
(α_i)	Link Twist Angle
(R_{ij})	Rotation from Frame i to Frame j
WC	Wrist Center
UR5	Universal Robots UR5 Manipulator
IRB	Industrial Robot by ABB
KR	KUKA Robot
FK (Symbolic)	Symbolic Forward Kinematics
FK (Numeric)	Numerical Forward Kinematics
IK (Analytical)	Analytical Inverse Kinematics
IK (Numeric)	Numerical Inverse Kinematics
RPY Angles	Orientation Representation Using Roll, Pitch, and Yaw
MATLAB	Matrix Laboratory
NumPy	Numerical Python Library
SymPy	Symbolic Python Library
SciPy	Scientific Python Library

VI. Introduction

A. Problem Background

Robotic manipulators are widely used in industrial automation, medical robotics, and advanced manufacturing due to their ability to perform precise and repetitive tasks. A manipulator with multiple degrees of freedom is capable of positioning and orienting its end-effector in three-dimensional space, enabling complex manipulation operations.

Kinematic modeling is a fundamental aspect of robot analysis and control. Forward kinematics determines the position and orientation of the end-effector given joint variables, while inverse kinematics computes the required joint variables for a desired end-effector pose. For manipulators with five and six degrees of freedom, inverse kinematics becomes particularly challenging due to the nonlinear and coupled nature of the governing equations. Accurate inverse kinematics solutions are essential for motion planning, trajectory generation, and real-time control of robotic systems.

B. Existing Solution Methods and Limitations

Several approaches have been proposed in the literature to solve the inverse kinematics problem of robotic manipulators. Numerical methods, such as iterative Jacobian-based techniques, are commonly used due to their simplicity and general applicability. However, these methods suffer from convergence issues, sensitivity to initial guesses, and high computational cost, especially in real-time applications [1].

Analytical inverse kinematics methods provide closed-form solutions that are computationally efficient and suitable for real-time control. Nevertheless, deriving analytical solutions for manipulators with higher degrees of freedom is mathematically complex and often restricted to specific manipulator geometries [2]. Hybrid approaches combine analytical and numerical techniques, but they may still face limitations related to singularities, joint limits, and multiple solution handling.

C. Motivation and Contributions of This Work

The motivation of this work is to develop a systematic and clear analytical framework for solving the forward and inverse kinematics of robotic manipulators with five and six degrees of freedom. This project focuses on deriving complete analytical solutions while explicitly addressing multiple inverse kinematics solutions, joint constraints, and singular configurations.

The main contributions of this work include:

1. Derivation of forward and inverse kinematics equations for three different robotic manipulators using the Denavit–Hartenberg convention.
2. Development of validated Python implementations for all derived kinematic models.
3. Design of a unified graphical user interface that integrates forward and inverse kinematics computations along with real-time two-dimensional and three-dimensional visualization.
4. Comprehensive validation of the analytical solutions through numerical testing and comparison between forward and inverse kinematics results.

D. Selected Industrial Manipulators

Manipulator 1: UR5 – Universal Robots

- Brief description:

The UR5 is a lightweight, six-degree-of-freedom (6-DOF) collaborative robotic manipulator developed by Universal Robots. It features a compact structure, high repeatability, and an intuitive control architecture, making it suitable for close human–robot interaction. The manipulator is characterized by a serial kinematic chain with revolute joints and a spherical wrist, which simplifies kinematic modeling and inverse kinematics derivation.

- Industrial applications:

The UR5 is widely used in industrial environments for tasks such as pick-and-place operations, assembly, machine tending, quality inspection, and packaging. Its collaborative nature allows it to operate safely alongside human workers without the need for extensive safety fencing.

- Reason for selection:

The UR5 was selected due to its extensive industrial adoption, well-documented kinematic structure, and suitability for analytical inverse kinematics derivation. Additionally, its standardized DH parameters and availability of CAD models make it an ideal candidate for validation of forward and inverse kinematics algorithms.

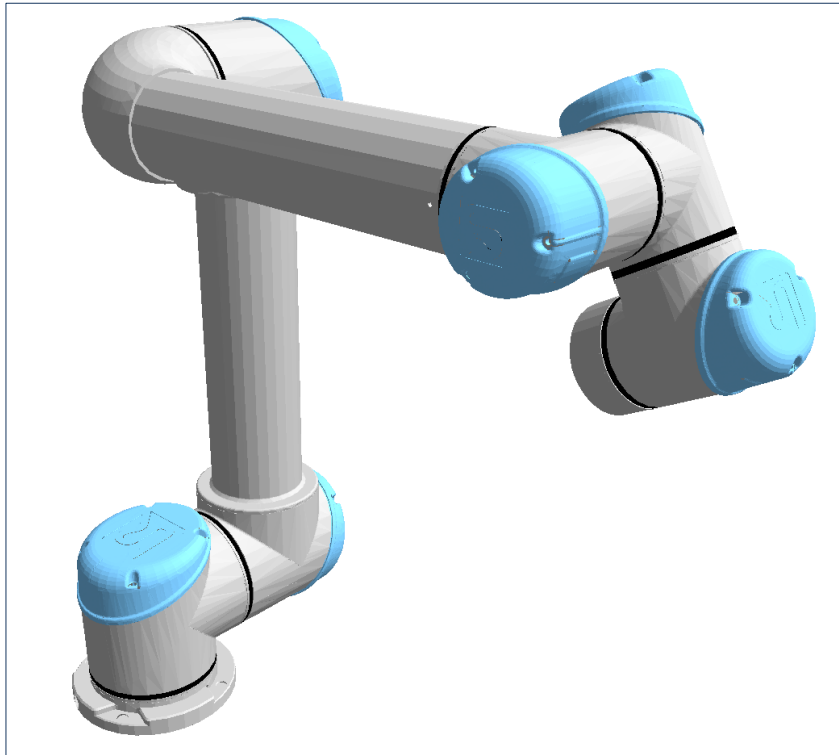


Figure 1: UR5 3D CAD Model

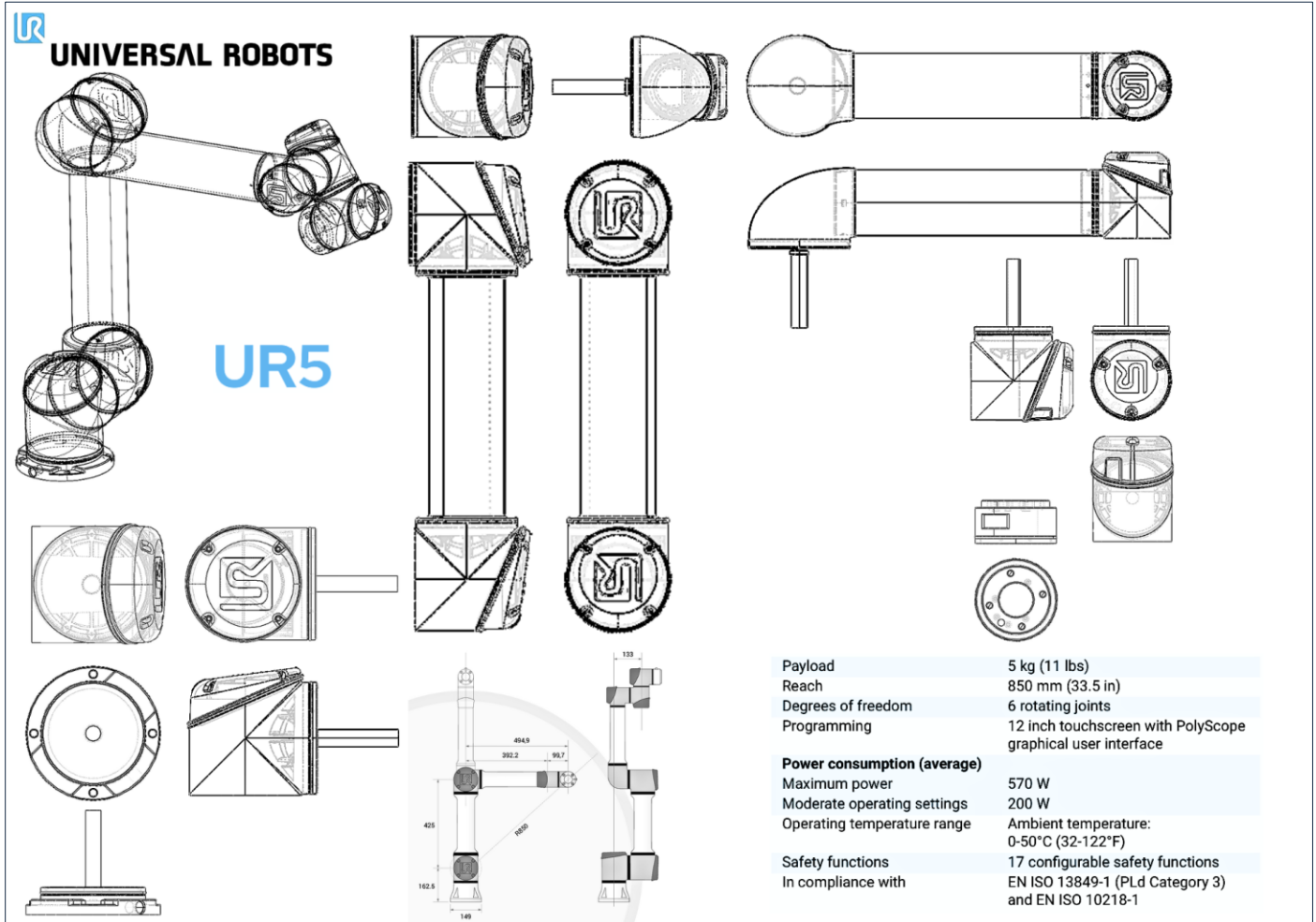


Figure 2: UR5 3D and 2D drawings including its parameters

• DH parameters

➤ DH Parameters Definition

Each link i of the UR5 robot is described using four DH parameters:

- θ_i : joint angle (variable)
- d_i : link offset (constant)
- a_i : link length (constant)
- α_i : link twist (constant)

➤ DH Table for UR5 manipulator:

Table 2 : DH Table for the manipulator UR5

Joint (i)	θ_i	d_i (m)	a_i (m)	α_i (deg)
1	θ_1	0.0892	0	90
2	θ_2	0	-0.425	0
3	θ_3	0	-0.3922	0
4	θ_4	0.1091	0	90
5	θ_5	0.0946	0	-90
6	θ_6	0.0823	0	0

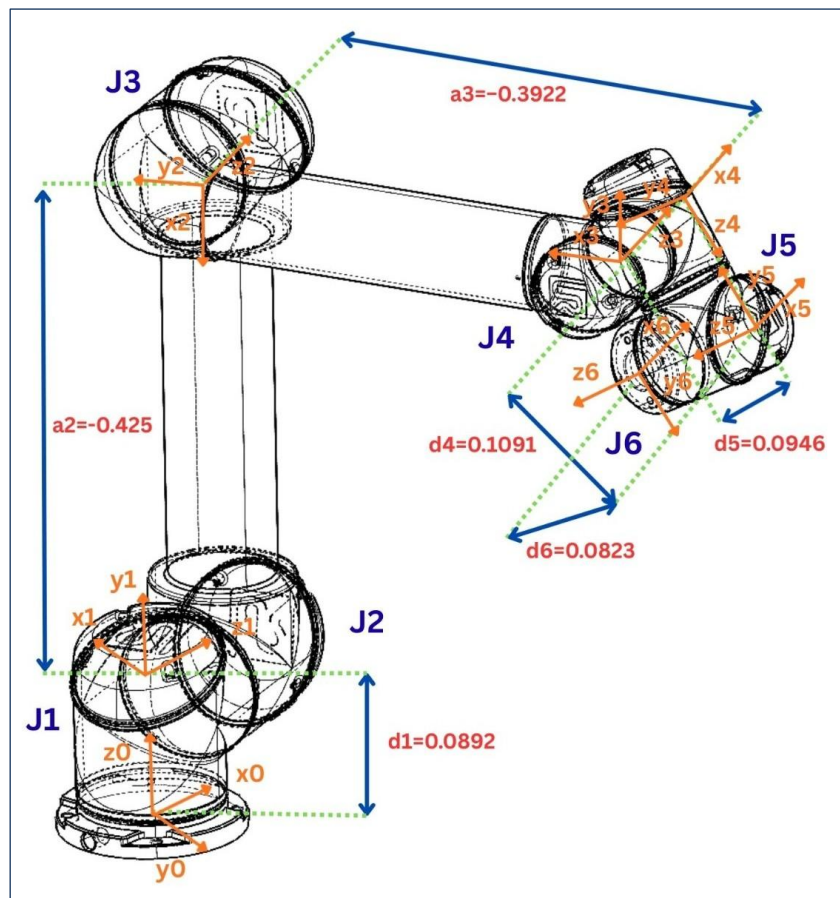


Figure 3: DH parameters showed on the manipulator UR5 itself showing its joint frames

➤ General DH Transformation Matrix (Fully Expanded)

The homogeneous transformation from frame $i - 1$ to frame i is:

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \dots\dots\dots \text{eq(1)}$$

This matrix simultaneously encodes:

- rotation about z_{i-1} by θ_i
- translation along z_{i-1} by d_i
- translation along x_i by a_i
- rotation about x_i by α_i

Manipulator 2: ABB IRB 1600

- Brief description:

The ABB IRB 1600 is a medium-payload, six-degree-of-freedom industrial robotic manipulator designed for high-speed and high-precision operations. It features a rigid mechanical structure, high torque joints, and an optimized serial-link configuration commonly used in industrial robotics.

- Industrial applications:

The IRB 1600 is extensively used in material handling, arc welding, machine loading and unloading, palletizing, and assembly operations. Its robust design enables reliable operation in demanding industrial environments such as automotive and manufacturing industries.

- Reason for selection:

This manipulator was chosen because it represents a classical industrial robot with a conventional serial architecture, making it suitable for rigorous analytical inverse kinematics derivation. Its widespread industrial usage and availability of manufacturer-provided kinematic data further support its selection.

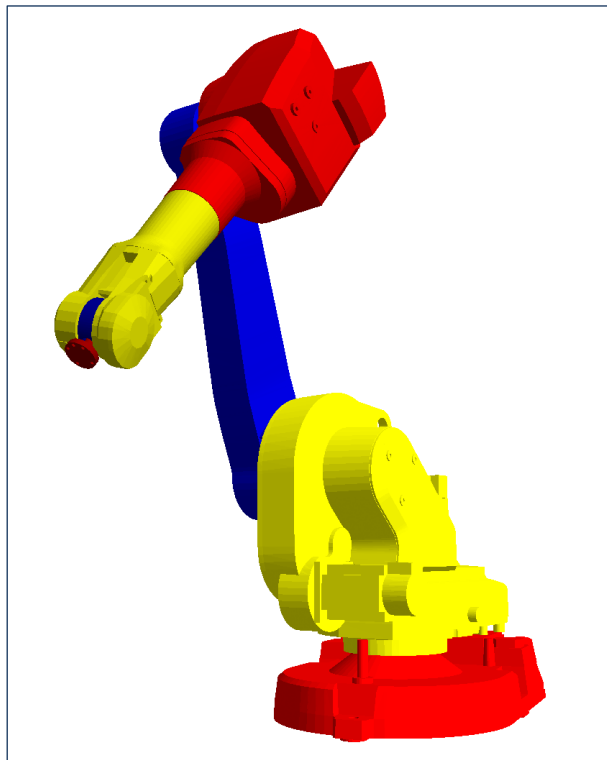


Figure 4: ABB IRB 1600 3D CAD Model

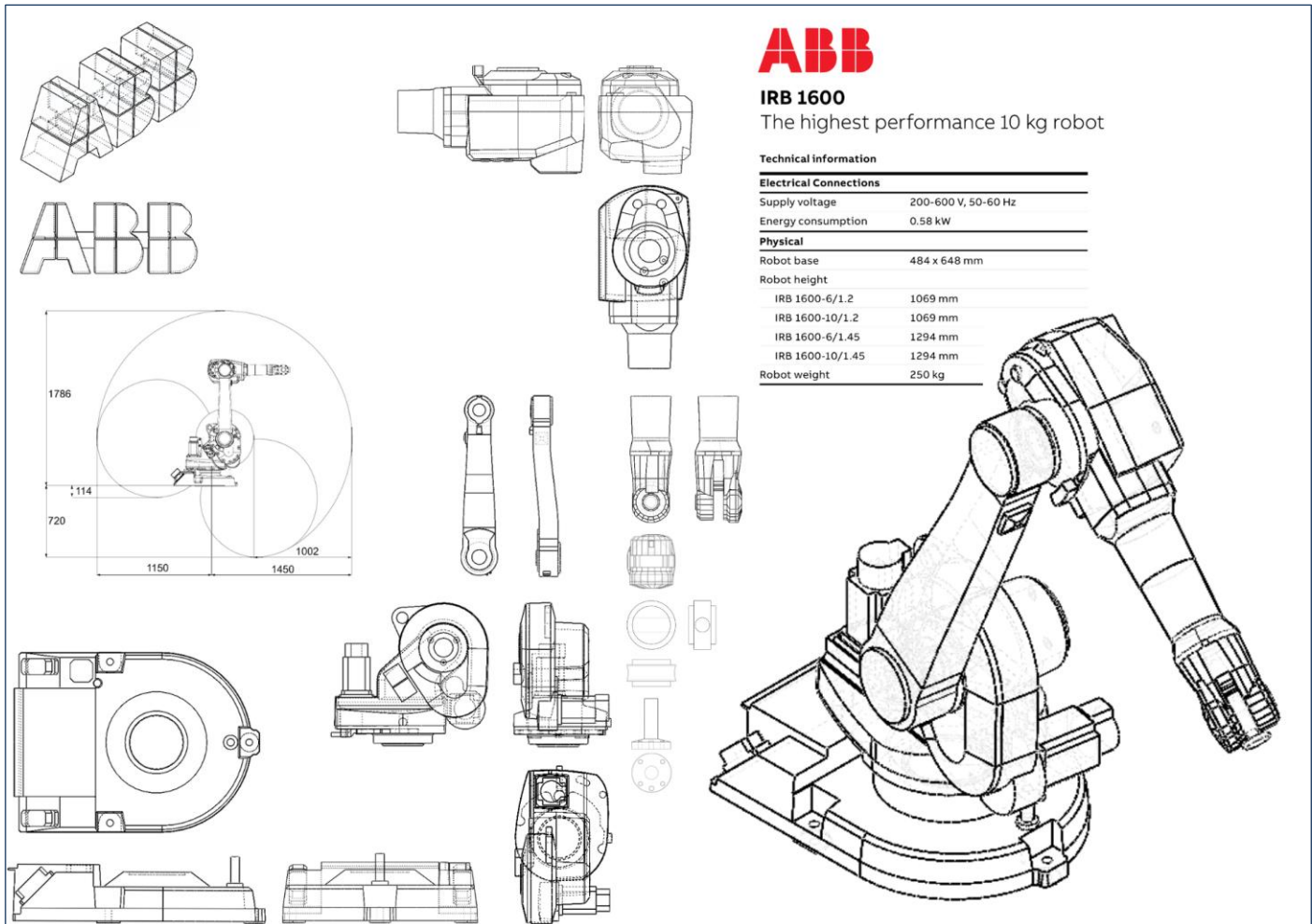


Figure 5 : ABB IRB 1600 3D and 2D drawings including its parameters

- **DH parameters**
- DH Table for ABB IRB 1600 manipulator:

Table 3: DH Table for ABB IRB 1600 manipulator

Joint (i)	θ_i	d_i (m)	a_i (m)	α_i (deg)
1	θ_1	0.72	0.448	-90
2	θ_2	0	1.066	0
3	θ_3	0	0.114	-90
4	θ_4	1.002	0	90
5	θ_5	0	0	-90
6	θ_6	0.25	0	0

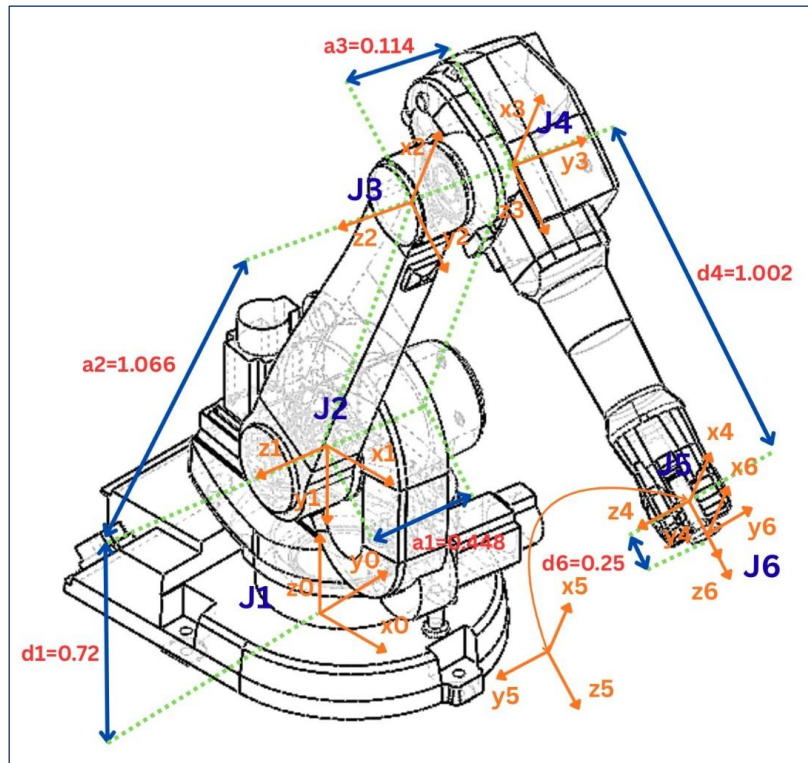


Figure 6: DH parameters showed on the manipulator ABB IRB 1600 itself showing its joint frames

➤ Denavit–Hartenberg Parameters of ABB IRB1600

The pose of the end effector relative to the base frame is given by:

$${}^0T_6 = A_1 A_2 A_3 A_4 A_5 A_6 \quad \dots\dots\dots \text{eq}(2)$$

where each A_i is computed using the general DH transformation matrix:

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \dots\dots\dots \text{eq}(3)$$

Manipulator 3: KUKA KR16

- Brief description:

The KUKA KR16 is a six-degree-of-freedom industrial robotic manipulator designed for precision tasks requiring moderate payload capacity. It features a compact mechanical design, high positioning accuracy, and a spherical wrist configuration, which is advantageous for orientation control.

- Industrial applications:

The KUKA KR16 is commonly employed in applications such as spot welding, assembly, material handling, packaging, and quality inspection. Its flexibility and precision make it suitable for a wide range of industrial automation tasks.

- Reason for selection:

The KUKA KR16 was selected due to its classical 6-DOF serial manipulator structure, which allows for closed-form inverse kinematics solutions. Its strong industrial relevance, along with accessible CAD models and technical documentation, makes it suitable for analytical modeling and validation.

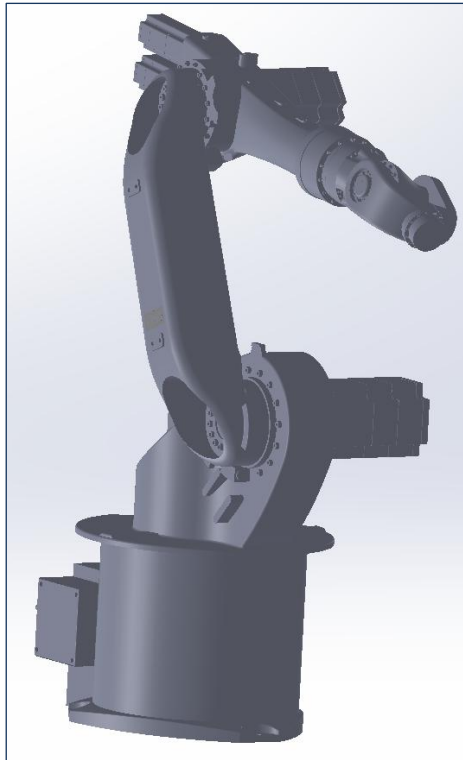


Figure 7: KUKA KR16 3D CAD Model

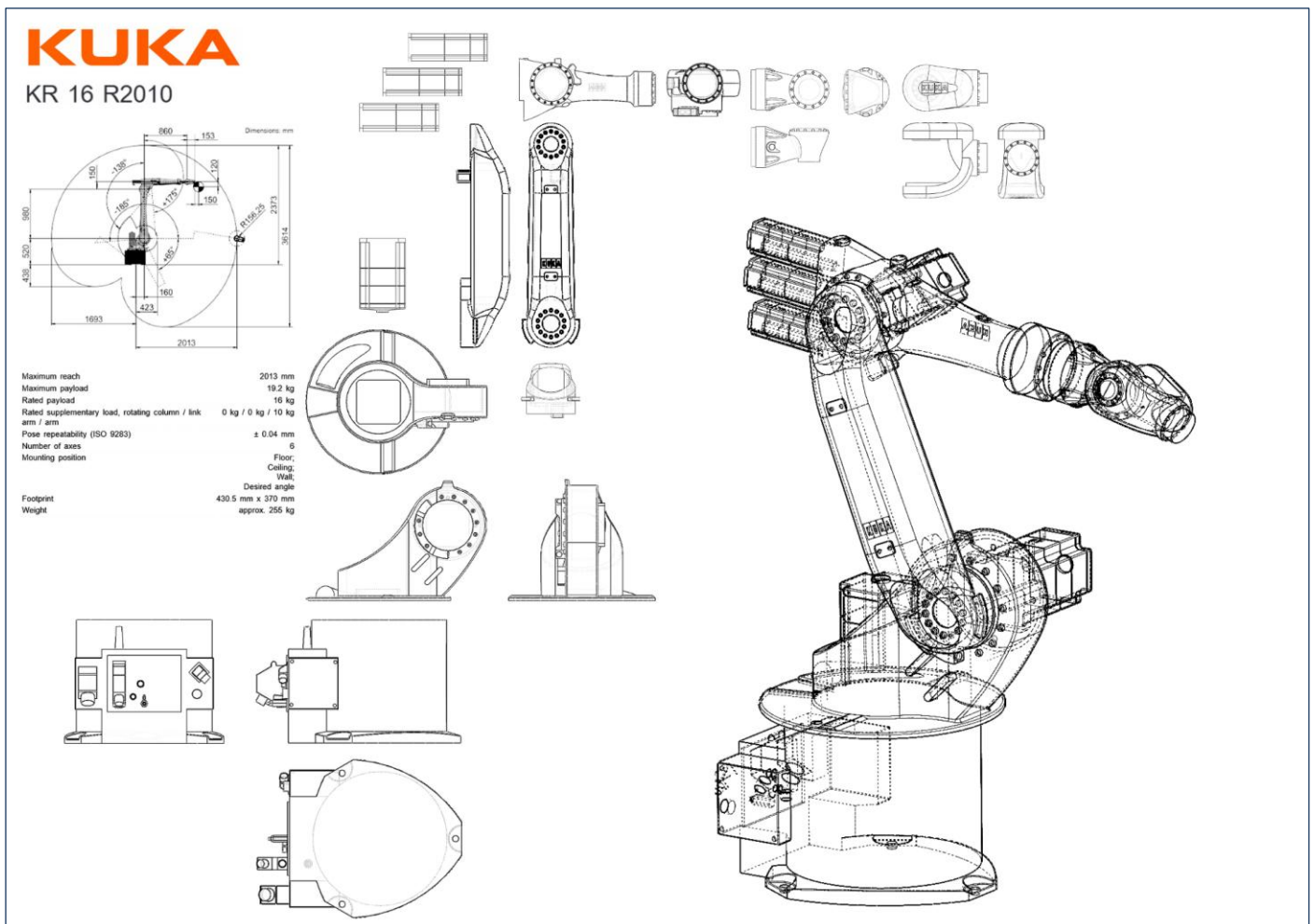


Figure 8: KUKA KR 16 3D and 2D drawings including its parameters

- DH parameters
- DH Table for KUKA KR16 manipulator:

Table 4: DH Table for KUKA KR16 manipulator

Joint (i)	θ_i	d_i (m)	a_i (m)	α_i (deg)
1	θ_1	0.675	0.26	90
2	θ_2	0	0.68	0
3	θ_3	0	0.035	90
4	θ_4	0.67	0	-90
5	θ_5	0	0	90
6	θ_6	0.115	0	0

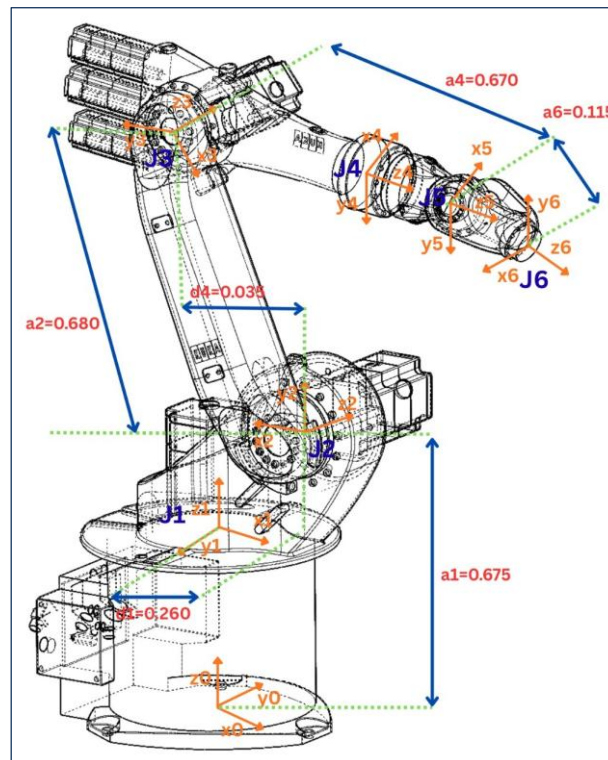


Figure 9: DH parameters showed on the manipulator KUKA KR16 itself showing its joint frames

➤ Denavit–Hartenberg Parameters for KUKA KR16:

The pose of the end effector relative to the base frame is given by:

$${}^0T_6 = A_1A_2A_3A_4A_5A_6 \quad \text{.....eq(4)}$$

where each A_i is computed using the general DH transformation matrix:

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{.....eq(5)}$$

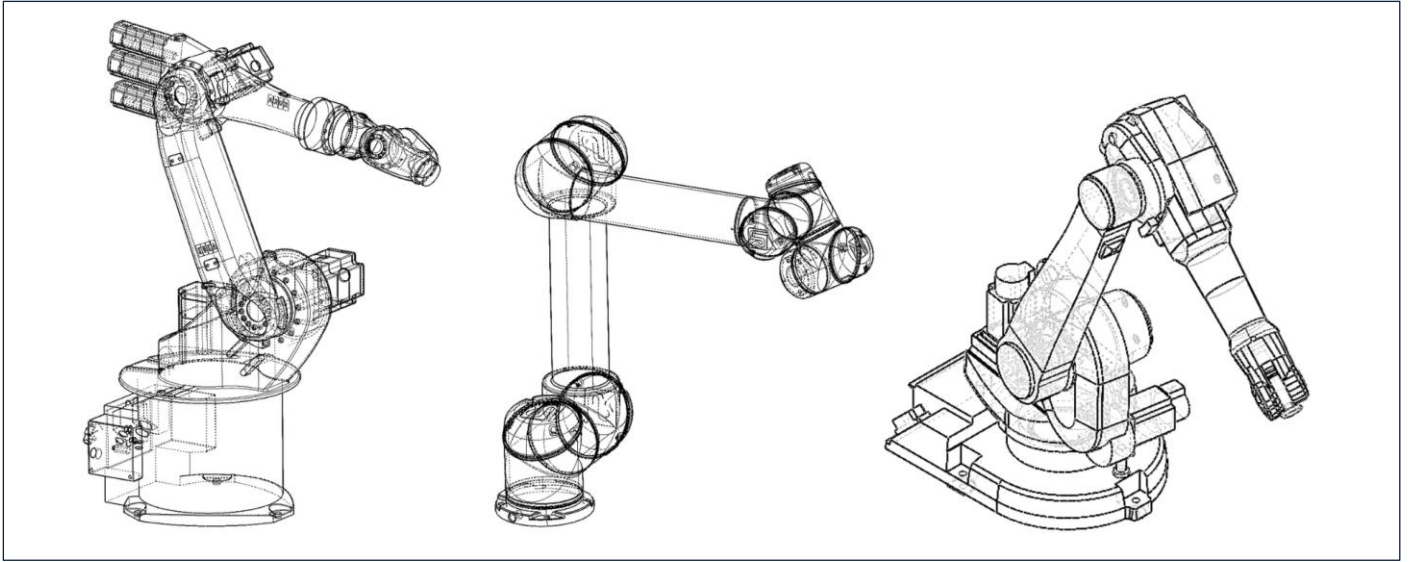


Figure 10 : 3D CAD Model for the three selected manipulators

VII. Problem Statement and Formulation

A. Problem Definition

The inverse kinematics problem of a robotic manipulator consists of determining the joint variables required to achieve a desired position and orientation of the end-effector. In this project, the problem is defined for three robotic manipulators with six degrees of freedom, each having a different geometric configuration.

Given a desired end-effector pose defined by its position and orientation in the Cartesian space, the objective is to compute all possible joint angle solutions that satisfy the kinematic constraints of the manipulator. Additionally, the derived solutions must respect joint limits and account for the existence of multiple inverse kinematics solutions and singular configurations.

B. Mathematical Formulation

The kinematic modeling of the manipulators is performed using the Denavit–Hartenberg (DH) convention. Each link is described by four DH parameters: link length a_i , link twist α_i , link offset d_i , and joint angle θ_i .

In the following section we are going to assume that :

$$c_i = \cos \theta_i \quad \text{.....eq(6)}$$

$$s_i = \sin \theta_i \quad \text{.....eq(7)}$$

1. Forward Kinematics Equation (General Form)

The pose of the end effector with respect to the base frame is obtained by:

$${}^0T_6 = A_1A_2A_3A_4A_5A_6 \quad \text{.....eq(8)}$$

1.1 We split it as:

$${}^0T_6 = \begin{bmatrix} R_{06} & p_{06} \\ 0 & 1 \end{bmatrix} \quad \text{.....eq(9)}$$

1.2 Individual Rotation Matrices:

i. Base Frame Rotation:

$$R_0 = R_{00} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \dots\dots\dots \text{eq(10)}$$

ii. Rotation from Frame 0 to Frame 1:

$$R_{01} = \begin{bmatrix} c_1 & -s_1 c_{\alpha_1} & s_1 s_{\alpha_1} \\ s_1 & c_1 c_{\alpha_1} & -c_1 s_{\alpha_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} \end{bmatrix} \dots\dots\dots \text{eq(11)}$$

iii. Rotation from Frame 1 to Frame 2:

$$R_{12} = \begin{bmatrix} c_2 & -s_2 c_{\alpha_2} & s_2 s_{\alpha_2} \\ s_2 & c_2 c_{\alpha_2} & -c_2 s_{\alpha_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} \end{bmatrix} \dots\dots\dots \text{eq(12)}$$

iv. Rotation from Frame 2 to Frame 3:

$$R_{23} = \begin{bmatrix} c_3 & -s_3 c_{\alpha_3} & s_3 s_{\alpha_3} \\ s_3 & c_3 c_{\alpha_3} & -c_3 s_{\alpha_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} \end{bmatrix} \dots\dots\dots \text{eq(13)}$$

v. Rotation from Frame 3 to Frame 4:

$$R_{34} = \begin{bmatrix} c_4 & -s_4 c_{\alpha_4} & s_4 s_{\alpha_4} \\ s_4 & c_4 c_{\alpha_4} & -c_4 s_{\alpha_4} \\ 0 & s_{\alpha_4} & c_{\alpha_4} \end{bmatrix} \dots\dots\dots \text{eq(14)}$$

vi. Rotation from Frame 4 to Frame 5:

$$R_{45} = \begin{bmatrix} c_5 & -s_5 c_{\alpha_5} & s_5 s_{\alpha_5} \\ s_5 & c_5 c_{\alpha_5} & -c_5 s_{\alpha_5} \\ 0 & s_{\alpha_5} & c_{\alpha_5} \end{bmatrix} \dots\dots\dots \text{eq(15)}$$

vii. Rotation from Frame 5 to Frame 6:

$$R_{56} = \begin{bmatrix} c_6 & -s_6 c_{\alpha_6} & s_6 s_{\alpha_6} \\ s_6 & c_6 c_{\alpha_6} & -c_6 s_{\alpha_6} \\ 0 & s_{\alpha_6} & c_{\alpha_6} \end{bmatrix} \dots\dots\dots \text{eq(16)}$$

1.3 Overall Rotation Matrix:

$$R_{06} = R_{01} R_{12} R_{23} R_{34} R_{45} R_{56} \dots\dots\dots \text{eq(17)}$$

1.4 Position Vector Equation

The end-effector position vector is:

$$p_{06} = \sum_{i=1}^6 \left(R_{0,i-1} \begin{bmatrix} a_i c_i \\ a_i s_i \\ d_i \end{bmatrix} \right) \dots\dots\dots \text{eq(18)}$$

Written term-by-term:

$$p_{06} = R_{01} \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ d_1 \end{bmatrix} + R_{02} \begin{bmatrix} a_2 c_2 \\ a_2 s_2 \\ d_2 \end{bmatrix} + R_{03} \begin{bmatrix} a_3 c_3 \\ a_3 s_3 \\ d_3 \end{bmatrix} + R_{04} \begin{bmatrix} a_4 c_4 \\ a_4 s_4 \\ d_4 \end{bmatrix} + R_{05} \begin{bmatrix} a_5 c_5 \\ a_5 s_5 \\ d_5 \end{bmatrix} + R_{06} \begin{bmatrix} a_6 c_6 \\ a_6 s_6 \\ d_6 \end{bmatrix} \dots\dots\dots \text{eq(19)}$$

where

$$p_{06} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \dots\dots\dots \text{eq(20)}$$

1.5 Final Forward Kinematics Matrix

$${}^0T_6 = \begin{bmatrix} R_{01}R_{12}R_{23}R_{34}R_{45}R_{56} & p_{06} \\ 000 & 1 \end{bmatrix} \dots\dots\dots \text{eq(21)}$$

1. Inverse Kinematics

A. ABB IRB1600 — Detailed Analytical IK

Given Pose

$${}^0T_6 = \begin{bmatrix} R_{06} & p_{06} \\ 0 & 1 \end{bmatrix}, p_{06} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \dots\dots\dots \text{eq(22)}$$

1. Wrist Center

$$p_{wc} = p_{06} - d_6 R_{06} \hat{z} \dots\dots\dots \text{eq(23)}$$

2. Base Joint (Two Solutions)

$$\theta_1^{(1)} = \text{atan } 2(p_{wc,y}, p_{wc,x}) \dots\dots\dots \text{eq(24)}$$

$$\theta_1^{(2)} = \theta_1^{(1)} + \pi \dots\dots\dots \text{eq(25)}$$

3. Coordinate Transformation

$${}^1p_{wc} = {}^0T_1^{-1} \begin{bmatrix} p_{wc} \\ 1 \end{bmatrix} \dots\dots\dots \text{eq(26)}$$

$$x_1 = ({}^1p_{wc})_x, y_1 = ({}^1p_{wc})_y \dots\dots\dots \text{eq(27)}$$

4. Link Geometry

$$L_2 = |a_2| \dots\dots\dots \text{eq(28)}$$

$$L_3 = \sqrt{a_3^2 + d_4^2} \dots\dots\dots \text{eq(29)}$$

$$\gamma = \text{atan } 2(d_4, a_3) \dots\dots\dots \text{eq(30)}$$

5. Elbow Joint (Two Solutions)

$$D = \frac{x_1^2 + y_1^2 - L_2^2 - L_3^2}{2L_2L_3} \dots\dots\dots \text{eq(31)}$$

$$\theta_3^{(1,2)} = \pm \arccos(D) \dots\dots\dots \text{eq(32)}$$

$$\therefore \theta_3^{(i)} = \theta_3'^{(i)} - \gamma$$

6. Shoulder Joint

$$\theta_2^{(i)} = \text{atan } 2(y_1, x_1) - \text{atan } 2(L_3 \sin \theta_3'^{(i)}, L_2 + L_3 \cos \theta_3'^{(i)}) \dots\dots\dots \text{eq(33)}$$

7. Wrist Rotation Matrix

$$R_{36} = R_{03}^T R_{06} \dots\dots\dots \text{eq(34)}$$

8. Wrist Pitch (Two Solutions)

$$\theta_5^{(1)} = \text{atan } 2\left(\sqrt{R_{36}(3,1)^2 + R_{36}(3,2)^2}, R_{36}(3,3)\right) \dots\dots\dots \text{eq(35)}$$

$$\theta_5^{(2)} = -\theta_5^{(1)} \dots\dots\dots \text{eq(36)}$$

9. Wrist Roll and Yaw

Non-Singular Case

$$\theta_4 = \text{atan } 2(R_{36}(2,3), R_{36}(1,3)) \dots\dots\dots \text{eq(37)}$$

$$\theta_6 = \text{atan } 2(-R_{36}(3,2), R_{36}(3,1)) \dots\dots\dots \text{eq(38)}$$

Singular Case ($\sin \theta_5 = 0$)

$$\theta_4 = \text{atan } 2(R_{36}(2,1), R_{36}(1,1)) \dots\dots\dots \text{eq(39)}$$

$$\theta_6 = 0 \dots\dots\dots \text{eq(40)}$$

DH Offset Correction

$$\theta_4 \leftarrow \theta_4 + \pi \dots\dots\dots \text{eq(41)}$$

B. KUKA KR16 — Detailed Analytical IK

1. Wrist Center

$$p_{wc} = p_{06} - d_6 R_{06} \hat{z} \dots\dots\dots \text{eq(42)}$$

2. Base Joint with Shoulder Offset

$$r_{xy} = \sqrt{p_{wc,x}^2 + p_{wc,y}^2} \dots\dots\dots \text{eq(43)}$$

$$\phi = \text{atan } 2(p_{wc,y}, p_{wc,x}) \dots\dots\dots \text{eq(44)}$$

$$\theta_1^{(1,2)} = \phi - \text{atan } 2\left(a_1, \pm \sqrt{r_{xy}^2 - a_1^2}\right) \dots\dots\dots \text{eq(45)}$$

3. Wrist Center in Frame 1

$${}^1p_{wc} = {}^0T_1^{-1}[p_{wc}, 1]^T \dots\dots\dots \text{eq(46)}$$

$$x_1 = ({}^1p_{wc})_x, z_1 = ({}^1p_{wc})_z \dots\dots\dots \text{eq(47)}$$

$$r = \sqrt{x_1^2 + z_1^2} \dots\dots\dots \text{eq(48)}$$

4. Geometry

$$L_2 = |a_2|, L_3 = \sqrt{a_3^2 + d_4^2} \dots\dots\dots \text{eq(49)}$$

$$\gamma = \text{atan } 2(d_4, a_3) \dots\dots\dots \text{eq(50)}$$

5. Elbow Joint

$$\theta_3'^{(1,2)} = \pm \arccos\left(\frac{r^2 - L_2^2 - L_3^2}{2L_2L_3}\right) \dots\dots\dots \text{eq(51)}$$

$$\theta_3^{(i)} = \theta_3'^{(i)} - \gamma \dots\dots\dots \text{eq(52)}$$

6. Shoulder Joint

$$\theta_2^{(i)} = \text{atan } 2(z_1, x_1) - \text{atan } 2(L_3 \sin \theta_3'^{(i)}, L_2 + L_3 \cos \theta_3'^{(i)}) \dots\dots\dots \text{eq(53)}$$

7. Wrist Rotation

$$R_{36} = R_{03}^T R_{06} \dots\dots\dots \text{eq(54)}$$

8. Wrist Joints

$$\theta_5^{(1,2)} = \pm \arccos(R_{36}(3,3)) \dots\dots\dots \text{eq(55)}$$

- Non-Singular

$$\theta_4 = \text{atan } 2 \left(\frac{R_{36}(2,3)}{\sin \theta_5}, \frac{R_{36}(1,3)}{\sin \theta_5} \right) \dots\dots\dots \text{eq}(56)$$

$$\theta_6 = \text{atan } 2 \left(\frac{R_{36}(3,2)}{\sin \theta_5}, \frac{-R_{36}(3,1)}{\sin \theta_5} \right) \dots\dots\dots \text{eq}(57)$$

- Singular

$$\theta_4 = \text{atan } 2(-R_{36}(2,1), R_{36}(1,1)), \theta_6 = 0 \dots\dots\dots \text{eq}(58)$$

C. UR5 — Fully Expanded Closed-Form IK

1. Wrist Center

$$p_{05} = p_{06} - d_6 R_{06} \hat{z} \dots\dots\dots \text{eq}(59)$$

$$r = \sqrt{p_{05,x}^2 + p_{05,y}^2} \dots\dots\dots \text{eq}(60)$$

2. Base Joint

$$\phi = \arccos \left(\frac{d_4}{r} \right) \dots\dots\dots \text{eq}(61)$$

$$\psi = \text{atan } 2(p_{05,y}, p_{05,x}) \dots\dots\dots \text{eq}(62)$$

$$\theta_1^{(1,2)} = \psi \pm \phi + \frac{\pi}{2} \dots\dots\dots \text{eq}(63)$$

3. Wrist Pitch

$$\cos \theta_5 = \frac{p_x \sin \theta_1 - p_y \cos \theta_1 - d_4}{d_6} \dots\dots\dots \text{eq}(64)$$

$$\theta_5^{(1,2)} = \pm \arccos(\cos \theta_5) \dots\dots\dots \text{eq}(65)$$

4. Wrist Yaw

$$\theta_6 = \text{atan } 2 \left(\frac{-R_{06}(1,2) \sin \theta_1 + R_{06}(2,2) \cos \theta_1}{\sin \theta_5}, \frac{R_{06}(1,1) \sin \theta_1 - R_{06}(2,1) \cos \theta_1}{\sin \theta_5} \right) \dots\dots\dots \text{eq}(66)$$

5. Intermediate Transform

$$T_{14} = T_{01}^{-1} T_{06} (T_{45} T_{56})^{-1} \dots\dots\dots \text{eq(67)}$$

$$x = T_{14,x}, y = T_{14,y} \dots\dots\dots \text{eq(68)}$$

6. Elbow Joint

$$\theta_3^{(1,2)} = \pm \arccos\left(\frac{x^2+y^2-a_2^2-a_3^2}{2a_2a_3}\right) \dots\dots\dots \text{eq(69)}$$

7. Shoulder Joint

$$\theta_2^{(i)} = \text{atan } 2(y, x) - \text{atan } 2\left(a_3 \sin \theta_3^{(i)}, a_2 + a_3 \cos \theta_3^{(i)}\right) \dots\dots\dots \text{eq(70)}$$

8. Wrist Roll

$$R_{34} = R_{03}^T R_{04} \dots\dots\dots \text{eq(71)}$$

$$\theta_4 = \text{atan } 2(R_{34}(2,1), R_{34}(1,1)) \dots\dots\dots \text{eq(72)}$$

9. Complete Solution Set

$$q^{(k)} = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6]^T \dots\dots\dots \text{eq(73)}$$

$$k = 1, \dots, 8 \dots\dots\dots \text{eq(74)}$$

C. Assumptions

The following assumptions are considered in this work:

1. All joints are assumed to be ideal revolute joints with no backlash or friction.
2. The links are considered rigid bodies with fixed dimensions.
3. The base of the manipulator is fixed in space.
4. Joint limits are predefined and respected during inverse kinematics solution selection.
5. External disturbances and dynamic effects are neglected, and only kinematic analysis is considered.

D. Algorithm Description

The forward kinematics algorithm computes the position and orientation of the end-effector given a set of joint configurations. The implementation follows a systematic transformation chain approach where individual link transformations are computed using the standard DH transformation matrix formula. For each joint, the algorithm constructs a four-by-four homogeneous transformation matrix that encodes both rotation and translation, calculated as a function of the four DH parameters. The transformation matrix employs trigonometric functions of the joint angle θ and link twist α , incorporating the link length a and offset d to fully describe the spatial relationship between consecutive coordinate frames

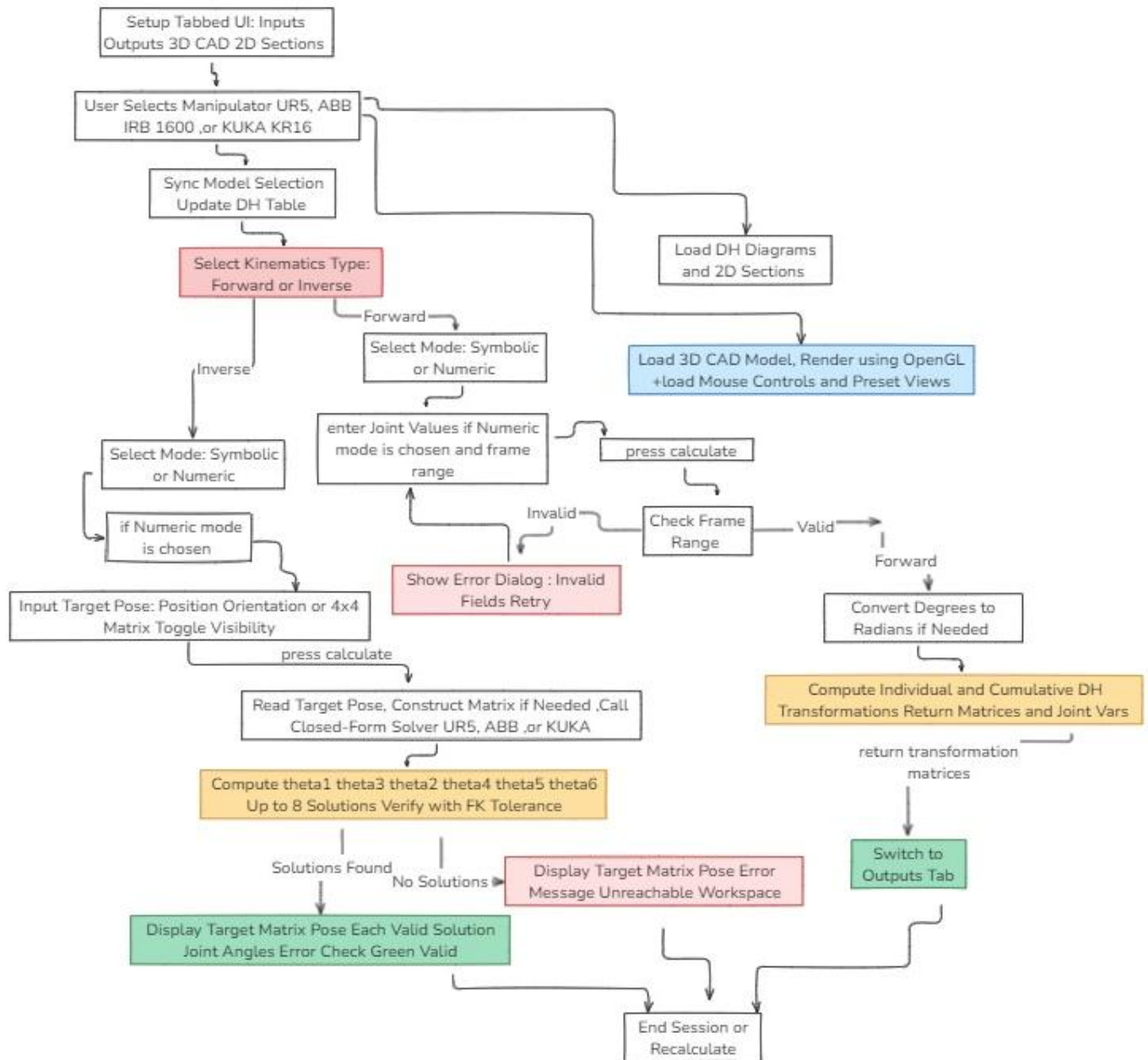


Figure 11 Flow chart for the Robotic Manipulator Kinematics Analysis Platform

The forward kinematics computation have two distinct modes: symbolic and numeric.

In the symbolic mode, the system utilizes SymPy library, using symbolic mathematics to generate closed-form expressions representing the kinematic relationships.

The algorithm creates symbolic variables for each joint angle and propagates these through the transformation chain, maintaining mathematical expressions rather than evaluating numerical values.

To address numerical noise inherent in floating-point arithmetic when displaying symbolic results, the system implements a cleaning function that removes coefficients smaller than a specified threshold , which equals $1 * 10^{-10}$, ensuring that spurious terms do not clutter the mathematical expressions presented to the user.

In numeric computation mode, the forward kinematics algorithm evaluates the transformation matrices using concrete joint angle values provided by the user. The computation begins by reading joint values from the user interface, converting angular measurements from degrees to radians when necessary based on the selected unit system. The algorithm then iterates through each joint, computing individual transformation matrices and maintaining both a cumulative transformation from the base frame to each successive joint frame and the individual transformations between adjacent frames.

The final end-effector pose is extracted from the cumulative transformation matrix by decomposing it into position coordinates and orientation angles using rotation matrix to Euler angle conversion through the Scipy spatial transform rotation module.

For the inverse kinematics, the platform implements closed-form analytical solutions specific to each manipulator, exploiting the spherical wrist configuration common to these industrial robots. The analytical approach leverages geometric decoupling, separating the positioning problem for the first three joints from the orientation problem for the wrist joints, thereby reducing the computational complexity and ensuring deterministic solution finding.

VIII. Solution Methodology

A. Parameters, Variables, and Coefficients

Table 5. Kinematic parameters and variables used in the analysis

Symbol	Description	Unit
(θ_i)	Joint angle of joint i	rad
(a_i)	Link length	m
(d_i)	Link offset	m
(α_i)	Link twist angle	rad
(T_0^n)	End-effector transformation matrix	—
$(\mathbf{q})$	Joint variable vector	—

Table 6 summarizes the kinematic parameters, variables, and coefficients used throughout the kinematic modeling and solution process.

B. Algorithm Implementation

The derived forward and inverse kinematics equations are implemented using Python to enable numerical evaluation and validation. Symbolic computation is initially employed to derive and simplify the kinematic equations, after which numerical substitution is used for simulation and visualization.

Separate functions are developed for forward kinematics and inverse kinematics for each manipulator. The forward kinematics function computes the end-effector pose based on given joint variables, while the inverse kinematics function calculates all possible joint angle solutions corresponding to a desired end-effector pose. Joint limit checks and solution validation procedures are incorporated to ensure physically feasible solutions.

A unified graphical user interface is designed to integrate all kinematic functions, allowing the user to select the manipulator, input joint variables or desired poses, and visualize the resulting configurations.

C. Software Implementation and Graphical User Interface

C.1 Input Interface

The developed graphical user interface (GUI) provides a structured and intuitive input environment that allows the user to define the manipulator configuration and select the desired kinematic computation. The input interface is designed to ensure flexibility, clarity, and consistency across all supported robotic manipulators.

The first input panel enables the user to select the target industrial manipulator from a predefined list, which includes the UR5, ABB IRB 1600, and KUKA KR16 robots. Once a manipulator is selected, the corresponding Denavit–Hartenberg (DH) parameters are automatically loaded into the system, ensuring that all subsequent computations are performed using the correct kinematic model.

The second panel allows the user to choose the type of kinematic analysis to be performed. The interface supports both Forward Kinematics (FK) and Inverse Kinematics (IK) computations. In FK mode, the joint variables are provided as inputs to compute the end-effector pose, whereas in IK mode, the desired end-effector pose is specified to determine the corresponding joint variables.

To enhance usability and numerical consistency, the GUI provides a computation mode selector that allows calculations to be carried out either symbolically or numerically. Additionally, the user may specify the unit system for joint variables, selecting between radians and degrees. This design choice ensures compatibility with both analytical derivations and numerical validation procedures.

The input interface also includes a frame selection module that allows the user to control which homogeneous transformation matrices are displayed. The user may choose to display all intermediate transformations or restrict the output to a specific range of frames, thereby facilitating step-by-step inspection of the kinematic chain.

For inverse kinematics analysis, two alternative methods of specifying the desired end-effector pose are supported. The pose can be entered either in terms of position and orientation parameters ($x,y,z,\alpha,\beta,\gamma$) or directly as a homogeneous transformation matrix. This dual input capability increases flexibility and allows the user to validate results using different representations of spatial pose.

Finally, the DH parameters table displayed within the input tab provides a clear summary of the kinematic parameters associated with each joint. When numerical computation is selected, the table allows direct entry of joint values, which are then used for real-time kinematic evaluation. This integrated design ensures transparency between theoretical modeling and computational implementation.

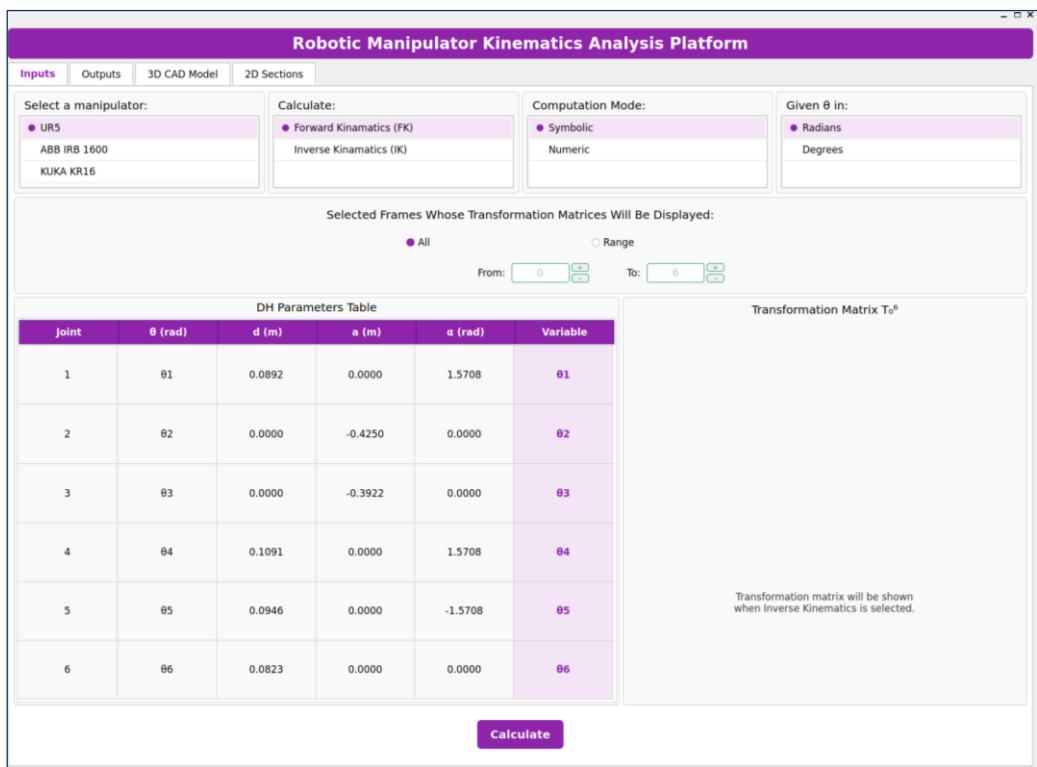


Figure 12: Graphical user interface showing the input tab for manipulator selection

C.2 Output Interface

The output interface of the developed application is designed to present the kinematic computation results in a structured, transparent, and interpretable manner. This interface provides both intermediate and final transformation matrices, allowing the user to trace the complete kinematic chain from the base frame to the end-effector.

For forward kinematics analysis, the output tab displays the symbolic homogeneous transformation matrices corresponding to each joint-to-joint frame transformation. These matrices are generated directly from the Denavit–Hartenberg parameters and are presented in symbolic form to preserve analytical clarity. The user may inspect individual transformation matrices T_i^{i+1} as well as cumulative transformations from the base frame T_0^i , enabling step-by-step verification of the kinematic formulation.

In addition to individual transformations, the interface highlights the overall forward kinematics transformation matrix T_0^6 , which represents the complete pose of the end-effector with respect to the base frame. This matrix is clearly separated from intermediate results to emphasize its role as the final output of the forward kinematics computation.

To improve readability and analytical insight, the output interface allows the transformation matrices to be viewed in different formats. The rotation submatrix (3×3), translation vector (3×1), and full homogeneous transformation matrix (4×4) can be displayed independently. This modular presentation facilitates focused analysis of orientation and position components without unnecessary visual clutter.

When inverse kinematics analysis is selected, the output tab presents the resulting joint variable solutions corresponding to the specified end-effector pose. The computed joint values are consistent with the selected unit system and computation mode. The interface supports numerical evaluation to validate analytical inverse kinematics solutions by substituting them back into the forward kinematics model.

Overall, the output interface serves as a validation and analysis tool that bridges theoretical kinematic derivations with computational implementation. By providing both symbolic expressions and numerical evaluations, the interface ensures correctness, traceability, and consistency across all supported robotic manipulators.

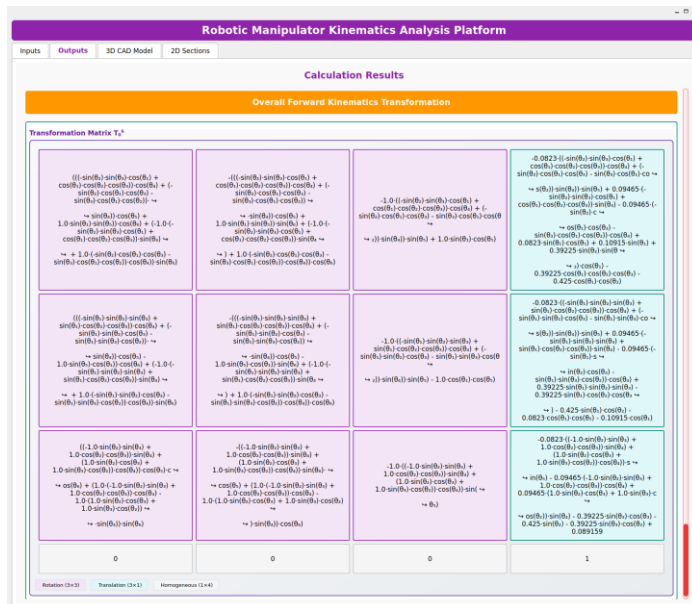


Figure 13 Output interface showing symbolic forward kinematics transformation for UR5

Robotic Manipulator Kinematics Analysis Platform

Inputs Outputs 3D CAD Model 2D Sections

Select a manipulator:
 UR5
 ABB IRB 1600
 • KUKA KR16

Calculate:
 Forward Kinematics (FK)
 • Inverse Kinematics (IK)

Computation Mode:
 • Symbolic
 Numeric

Given θ in:
 • Radians
 Degrees

Select the method of entering the desired End-Effector Pose
☒ Position_Orientation (x, y, z, α , β , γ) ☐ Homogeneous Transformation Matrix (4x4)

α (Roll) β (Pitch) γ (Yaw)
 x y z

Joint	θ (rad)	d (m)	a (m)	α (rad)	Variable
1	θ_1	0.6750	0.2600	1.5708	θ_1
2	θ_2	0.0000	0.6800	0.0000	θ_2
3	θ_3	0.0000	0.0350	1.5708	θ_3
4	θ_4	0.6700	0.0000	-1.5708	θ_4
5	θ_5	0.0000	0.0000	1.5708	θ_5
6	θ_6	0.1150	0.0000	0.0000	θ_6

Transformation Matrix T_n^0

Transformation matrix will be shown when Inverse Kinematics is selected.

Calculate

Figure 14 Robotic Manipulator Kinematics Analysis Platform – Input Interface

Robotic Manipulator Kinematics Analysis Platform

Inputs **Outputs** 3D CAD Model 2D Sections

Calculation Results

Inverse Kinematics - Symbolic Derivation

Target End-Effector Pose (Symbolic)

Symbolic Variables

x	X Position	y	Y Position	z	Z Position
α	Roll Angle	β	Pitch Angle	γ	Yaw Angle

Target Transformation Matrix T_n^0

$\cos(\beta) \cos(\gamma)$	$\sin(\alpha) \sin(\beta) \cos(\gamma) - \sin(\gamma) \cos(\alpha)$	$\sin(\alpha) \sin(\gamma) + \sin(\beta) \cos(\alpha) \cos(\gamma)$	x
$\sin(\gamma) \cos(\beta)$	$\sin(\alpha) \sin(\beta) \sin(\gamma) + \cos(\alpha) \cos(\gamma)$	$-\sin(\alpha) \cos(\gamma) + \sin(\beta) \sin(\gamma) \cos(\alpha)$	y
$-\sin(\beta)$	$\sin(\alpha) \cos(\beta)$	$\cos(\alpha) \cos(\beta)$	z
0	0	0	1

Rotation (3x3) Translation (3x1) Homogeneous (4x4)

Solution Derivation Steps

1 Computing Wrist Center (p_w)

The wrist center is computed by offsetting from the target position along the approach vector:

$p_{wx} = x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma)$
 $p_{wy} = y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha)$
 $p_{wz} = z - 0.115 \cos(\alpha) \cos(\beta)$

2 Solving for θ_1 (Base Rotation)

Two solutions exist due to geometric ambiguity:

$\theta_1^{(1)} = \text{atan2}(0.26, \sqrt{(x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))^2 + (y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha))^2 - 0.0676}) + \text{atan2}(y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha), x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))$
 $\theta_1^{(2)} = \text{atan2}(0.26, -\sqrt{(x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))^2 + (y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha))^2 - 0.0676}) + \text{atan2}(y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha), x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))$

3 Solving for θ_2 (Wrist Rotation)

Computed from the geometric constraint:

$\theta_2^{(1)} = \text{acos}(R_{36_22})$
 $\theta_2^{(2)} = -\text{acos}(R_{36_22})$

4 Solving for θ_3 (Wrist Orientation)

Derived from rotation matrix elements:

Figure 15 Symbolic Inverse Kinematics Target End-Effector Pose Representation

Robotic Manipulator Kinematics Analysis Platform

Inputs **Outputs** 3D CAD Model 2D Sections

Calculation Results

1 Computing Wrist Center (p_w)

The wrist center is computed by offsetting from the target position along the approach vector:

$p_{wx} = x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma)$
 $p_{wy} = y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha)$
 $p_{wz} = z - 0.115 \cos(\alpha) \cos(\beta)$

2 Solving for θ_1 (Base Rotation)

Two solutions exist due to geometric ambiguity:

$\theta_1^{(1)} = \text{atan2}(0.26, \sqrt{(x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))^2 + (y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha))^2 - 0.0676}) + \text{atan2}(y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha), x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))$
 $\theta_1^{(2)} = \text{atan2}(0.26, -\sqrt{(x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))^2 + (y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha))^2 - 0.0676}) + \text{atan2}(y + 0.115 \sin(\alpha) \cos(\gamma) - 0.115 \sin(\beta) \sin(\gamma) \cos(\alpha), x - 0.115 \sin(\alpha) \sin(\gamma) - 0.115 \sin(\beta) \cos(\alpha) \cos(\gamma))$

3 Solving for θ_2 (Wrist Rotation)

Computed from the geometric constraint:

$\theta_2^{(1)} = \text{acos}(R_{36_22})$
 $\theta_2^{(2)} = -\text{acos}(R_{36_22})$

4 Solving for θ_3 (Wrist Orientation)

Derived from rotation matrix elements:

Figure 16 Inverse Kinematics Solution Steps: Wrist Center and Base Joint Computation

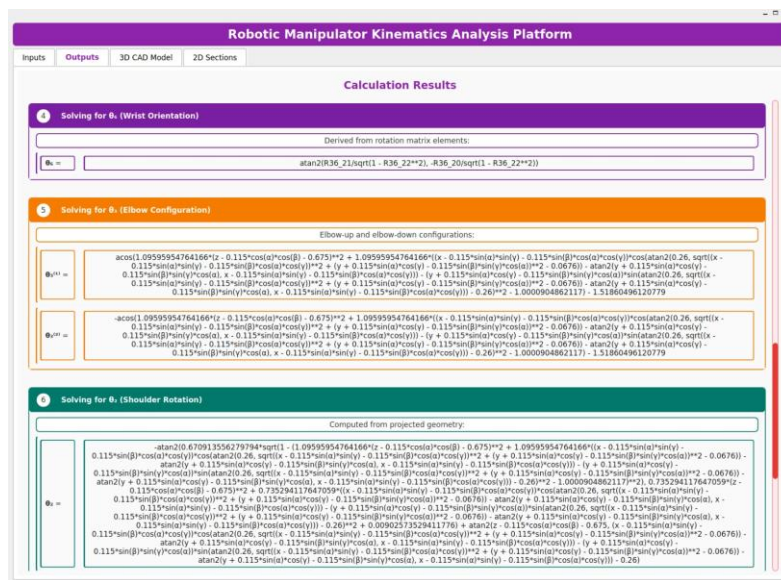


Figure 17 Inverse Kinematics Joint Angle Derivation and Configuration Analysis

C.3 CAD Visualization Module

To complement the numerical and analytical kinematic analysis, the developed application includes two dedicated CAD model visualization tabs as shown in Figures 18 and 19. These tabs provides a three-dimensional and two-dimensional representation of the selected robotic manipulator, enabling visual inspection of the robot structure and its spatial configuration.

The first tab, “3D CAD Model”, presents an interactive three-dimensional visualization of the selected robot, including the UR5, ABB IRB 1600, and KUKA KR16 manipulators. This view allows the user to rotate, zoom, and inspect the robot from different perspectives, as well as switch between predefined views such as isometric, front, top, and side views. The displayed models are consistent with the kinematic structure defined by the Denavit–Hartenberg parameters used in the analytical formulation. This consistency ensures that the visual representation accurately reflects the underlying kinematic model.

The primary purpose of the CAD visualization is to enhance the interpretability of the kinematic results by providing intuitive spatial context. By observing the manipulator geometry, the user can better understand joint orientations, link arrangements, and the overall configuration of the robot. This is particularly useful when validating forward and inverse kinematics results, as the visual pose of the end-effector can be compared with the computed position and orientation.

The second tab, “2D Sections,” provides detailed two-dimensional technical views of the selected manipulator, including orthographic projections, sectional drawings, and dimensional layouts derived from the robot’s mechanical design. These views highlight key geometric features such as link lengths, joint locations, workspace envelopes, and mounting dimensions. Unlike the 3D visualization, which emphasizes spatial intuition, the 2D section views focus on geometric clarity and structural understanding.

The 2D visualization serves both educational and verification purposes. It allows users to relate the abstract kinematic parameters, such as link lengths and offsets, to their actual physical dimensions, reinforcing the connection between the analytical model and the real robot structure.

Although the CAD model visualization does not perform computations directly, it serves as an important validation and educational tool. By integrating analytical results with a three-dimensional representation, the application bridges the gap between mathematical modeling and physical intuition, thereby improving the overall usability and clarity of the kinematic analysis platform.

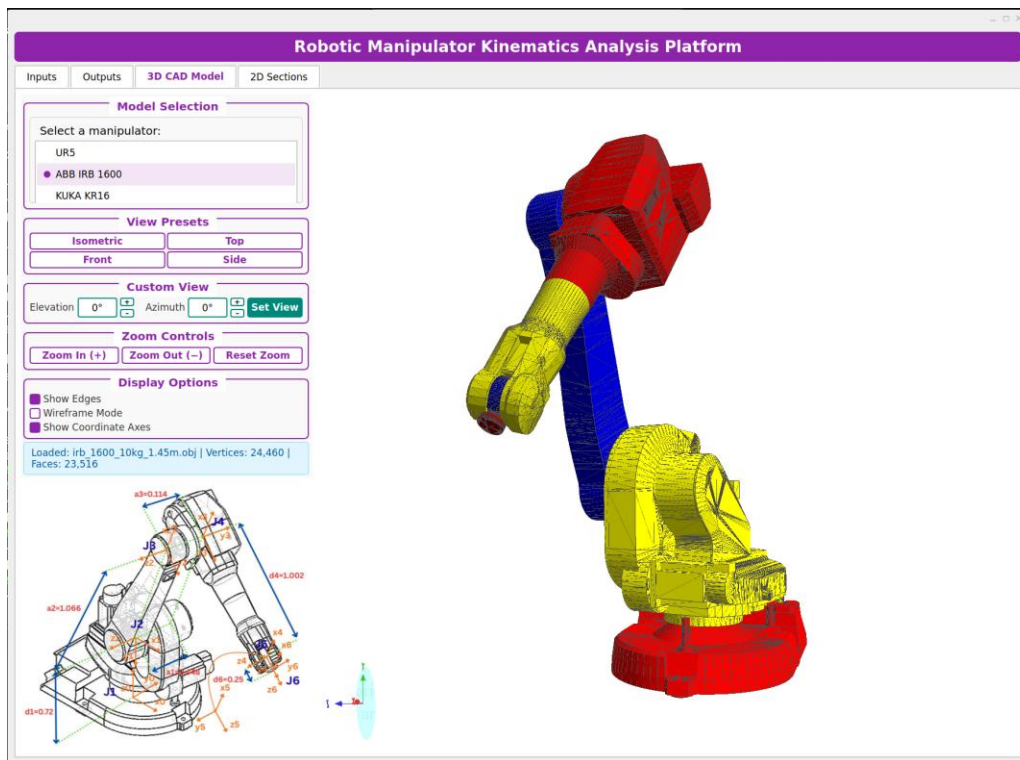


Figure 18 : 3D CAD model visualization tab showing the three-dimensional representation

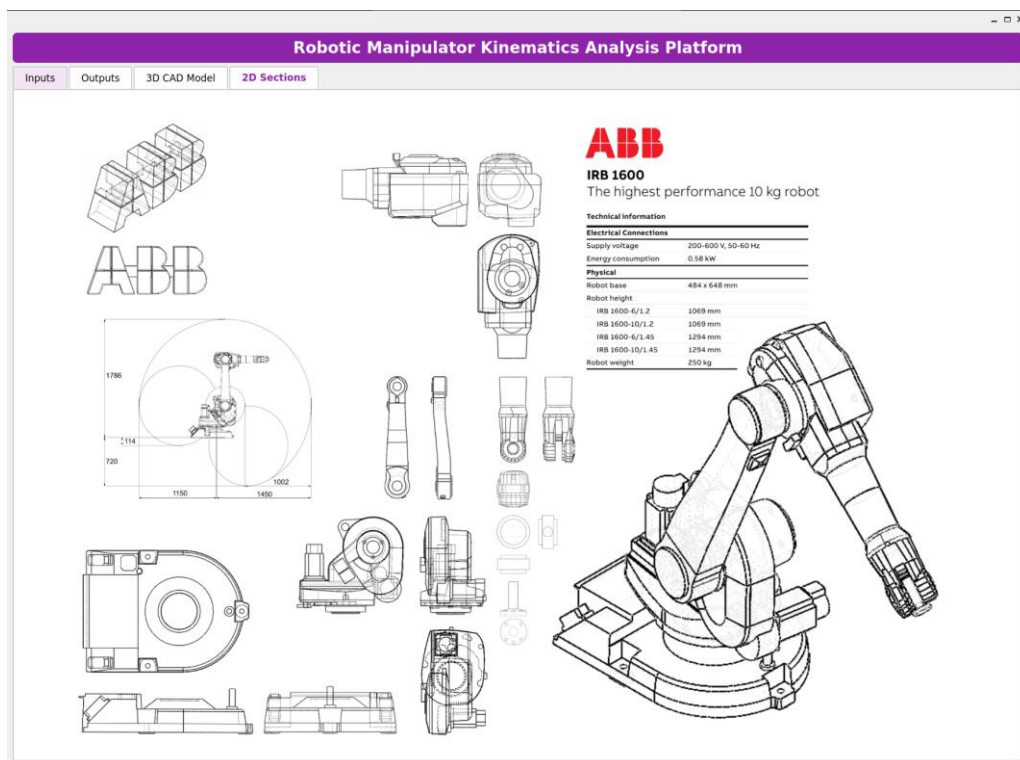


Figure 19: 2D Sections tab showing different 2d views , projections and some manufacturer details

IX. Results and Discussion

A. Numerical Examples and Results

Manipulator 1: UR5 – Universal Robots

Case 1: Forward Kinematics

$$q_{in}=[0.300000, -1.000000, 1.200000, -0.700000, 1.000000, 0.400000] \text{ rad}$$

Convert to degrees:

- $\theta_1 = 0.3 \text{ rad} = 17.19^\circ$
- $\theta_2 = -1.0 \text{ rad} = -57.30^\circ$
- $\theta_3 = 1.2 \text{ rad} = 68.75^\circ$
- $\theta_4 = -0.7 \text{ rad} = -40.11^\circ$
- $\theta_5 = 1.0 \text{ rad} = 57.30^\circ$
- $\theta_6 = 0.4 \text{ rad} = 22.92^\circ$

Joint 1 Transformation A_1

Given (UR5 standard DH):

- $\alpha_1 = +90^\circ, a_1 = 0, d_1 = 0.089159\text{m}$
- $\theta_1 = 0.3\text{rad}$

Trig:

- $\cos \theta_1 = \cos(0.3) = 0.955336$
- $\sin \theta_1 = \sin(0.3) = 0.295520$
- $\cos \alpha_1 = 0, \sin \alpha_1 = 1$

Substitute into DH matrix:

$$A_1 = \begin{bmatrix} 0.955336 & 0 & 0.295520 & 0 \\ 0.295520 & 0 & -0.955336 & 0 \\ 0 & 1 & 0 & 0.089159 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 2 Transformation A_2

Given:

- $\alpha_2 = 0^\circ, a_2 = -0.425\text{m}, d_2 = 0$
- $\theta_2 = -1.0\text{rad}$

Trig:

- $\cos \theta_2 = \cos(-1.0) = 0.540302$
- $\sin \theta_2 = \sin(-1.0) = -0.841471$

So:

$$A_2 = \begin{bmatrix} 0.540302 & 0.841471 & 0 & -0.229628 \\ -0.841471 & 0.540302 & 0 & 0.357625 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Joint 3 Transformation A_3

Given:

- $\alpha_3 = 0^\circ$, $a_3 = -0.39225\text{m}$, $d_3 = 0$
- $\theta_3 = 1.2\text{rad}$

Trig:

- $\cos \theta_3 = \cos(1.2) = 0.362358$
- $\sin \theta_3 = \sin(1.2) = 0.932039$

So:

$$A_3 = \begin{bmatrix} 0.362358 & -0.932039 & 0 & -0.142128 \\ 0.932039 & 0.362358 & 0 & -0.365732 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

After multiplying:

$${}^0T_6 = A_1 A_2 A_3 A_4 A_5 A_6 \dots\dots\dots \text{eq(75)}$$

The final end-effector pose (from your file) is:

$${}^0T_6 = \begin{bmatrix} 0.824625 & 0.148621 & -0.545808 & -0.642648 \\ -0.556194 & 0.388978 & -0.734400 & -0.359593 \\ 0.103160 & 0.909180 & 0.403423 & 0.318995 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Thus:

- $x = -0.642648\text{m}$, $y = -0.359593\text{m}$, $z = 0.318995\text{m}$

Entering the given inputs in our GUI's input tab as shown in figure 20:

Robotic Manipulator Kinematics Analysis Platform

Inputs

Outputs

3D CAD Model

2D Sections

Select a manipulator:

UR5

ABB IRB 1600

KUKA KR16

Calculate:

Forward Kinematics (FK)

Inverse Kinematics (IK)

Computation Mode:

Symbolic

Numeric

Given θ in:

Radians

Degrees

Selected Frames Whose Transformation Matrices Will Be Displayed:

All

Range

From:

0

To:

6

DH Parameters Table

Joint	θ (rad)	d (m)	a (m)	α (rad)	Variable	Value
1	θ_1	0.0892	0.0000	1.5708	θ_1	0.3
2	θ_2	0.0000	-0.4250	0.0000	θ_2	-1
3	θ_3	0.0000	-0.3922	0.0000	θ_3	1.2
4	θ_4	0.1091	0.0000	1.5708	θ_4	-0.7
5	θ_5	0.0946	0.0000	-1.5708	θ_5	1
6	θ_6	0.0823	0.0000	0.0000	θ_6	0.4

Transformation Matrix T_0^6

Transformation matrix will be shown when Inverse Kinematics is selected.

Calculate

Figure 20: Entering case 1 inputs on the designed Platform

We got the following results in figure 21 and 22:

Robotic Manipulator Kinematics Analysis Platform

Inputs

Outputs

3D CAD Model

2D Sections

Calculation Results

Forward Kinematics Results (Numeric)

Input Joint Values

θ_1	0.300000 rad	θ_2	-1.000000 rad	θ_3	1.200000 rad
θ_4	-0.700000 rad	θ_5	1.000000 rad	θ_6	0.400000 rad

Individual Link Transformations

Transformation Matrix T_0^1

0.955336	-0.000000	0.295520	0.000000
0.295520	0.000000	-0.955336	0.000000
0.000000	1.000000	0.000000	0.089159
0.000000	0.000000	0.000000	1.000000

Rotation | Translation | Homogeneous

Transformation Matrix T_1^2

0.540302	0.841471	-0.000000	-0.229628
-0.841471	0.540302	-0.000000	0.357625
0.000000	0.000000	1.000000	0.000000
0.000000	0.000000	0.000000	1.000000

Rotation | Translation | Homogeneous

Transformation Matrix T_2^3

0.362358	-0.932039	0.000000	-0.142135
0.932039	0.362358	-0.000000	-0.365592
0.000000	0.000000	1.000000	0.000000
0.000000	0.000000	0.000000	1.000000

Rotation | Translation | Homogeneous

Figure 21: The GUI's results for case 1 showing the input joint values and individual link transformation

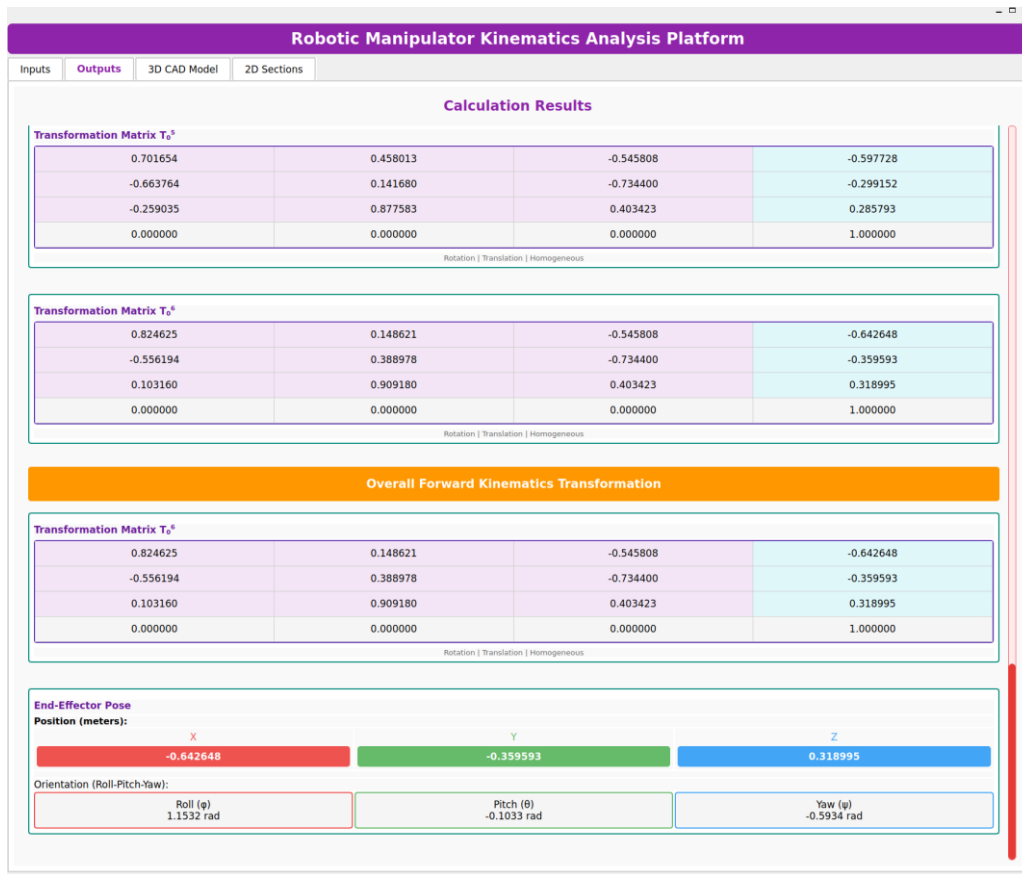


Figure 22: The GUI's results for case 1 showing the output over all transformation and the end-effector pose

Which shows an exact match between the manual calculations and the ones provided by the GUI.

Case 2: Inverse Kinematics

Given the desired pose:

$$T_0^6 = \begin{bmatrix} R_{06} & p_{06} \\ 0 & 1 \end{bmatrix}, p_{06} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \dots\dots\dots \text{eq(76)}$$

Goal: determine $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$.

Wrist Center Calculation :

UR5 computes the wrist center by removing the last link offset:

$$p_{05} = p_{06} - d_6 \hat{z}_{06} \quad \dots\dots\dots \text{eq(77)}$$

where:

$$\hat{z}_{06} = \begin{bmatrix} R_{06}(1,3) \\ R_{06}(2,3) \\ R_{06}(3,3) \end{bmatrix} \quad \dots\dots\dots \text{eq(78)}$$

Using the Case 1 matrix:

$$p_{06} = [-0.642648, -0.359593, 0.318995]^T$$

$$\hat{z}_{06} = [-0.545808, -0.734400, 0.403423]^T$$

$$d_6 = 0.0823$$

So:

$$p_{05} = p_{06} - 0.0823\hat{z}_{06} \quad \dots\dots\dots\text{eq(79)}$$

Compute each component:

- $x_{05} = -0.642648 - 0.0823(-0.545808) = -0.597728$
- $y_{05} = -0.359593 - 0.0823(-0.734400) = -0.299152$
- $z_{05} = 0.318995 - 0.0823(0.403423) = 0.285793$

So:

$$p_{05} = [-0.597728, -0.299152, 0.285793]^T$$

Base Joint Angle θ_1

UR5 uses:

$$r = \sqrt{x_{05}^2 + y_{05}^2}, \phi = \cos^{-1}\left(\frac{d_4}{r}\right), \psi = \text{atan2}(y_{05}, x_{05}) \quad \dots\dots\dots\text{eq(80)}$$

$$\theta_1 = \psi \pm \phi + \frac{\pi}{2} \quad \dots\dots\dots\text{eq(81)}$$

(with 2 base solutions)

Numbers:

- $r = \sqrt{(-0.597728)^2 + (-0.299152)^2} = 0.668409$
- $d_4 = 0.10915 \Rightarrow \frac{d_4}{r} = 0.163298$
- $\phi = \cos^{-1}(0.163298) = 1.406764$
- $\psi = \text{atan2}(-0.299152, -0.597728) = -2.677560$

So:

- $\theta_1 = \psi + \phi + \frac{\pi}{2} = 0.300000 \text{ rad (valid branch)}$
(Second branch also exists, but the solver picks the consistent one.)

Shoulder-Elbow Geometry :

Define:

$$r = \sqrt{x_{wc}^2 + y_{wc}^2} \quad \dots\dots\dots\text{eq(82)}$$

$$s = z_{wc} - d_1 \quad \dots\dots\dots\text{eq(83)}$$

Apply the cosine law:

$$D = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2a_3} \quad \dots\dots\dots\text{eq(84)}$$

$$\theta_3 = \pm \cos^{-1}(D) \dots\dots\dots \text{eq(85)}$$

Shoulder Angle θ_2 :

UR5 uses:

$$\theta_2 = \text{atan2}(y, x) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 \cos \theta_3) \dots\dots\dots \text{eq(86)}$$

Substituting:

$$\theta_2 = -1.000000 \text{ rad}$$

Wrist Angles $\theta_4, \theta_5, \theta_6$:

After $\theta_1, \theta_2, \theta_3$ are found, the remaining wrist is extracted from the remaining rotation (same idea as:

$R_{36} = R_{03}^T R_{06}$), producing:

$$[\theta_4, \theta_5, \theta_6] = [-0.700000, 1.000000, 0.400000] \text{ rad}$$

Final IK Output for UR5 Case 2:

$$q_{ik} = [0.300000, -1.000000, 1.200000, -0.700000, 1.000000, 0.400000]$$

This matches the forward kinematics input exactly, and by the consistency provided the methods used are verified.

We also notice that UR5 inverse kinematics is solved in two main parts:

1. Position part (solve $\theta_1, \theta_2, \theta_3$) using a wrist-center style decoupling
2. Orientation part (solve $\theta_4, \theta_5, \theta_6$) using the remaining rotation matrix

Using our platform we see that it gave us 8 solutions, one of them is the set of joints value we started with in case 1.

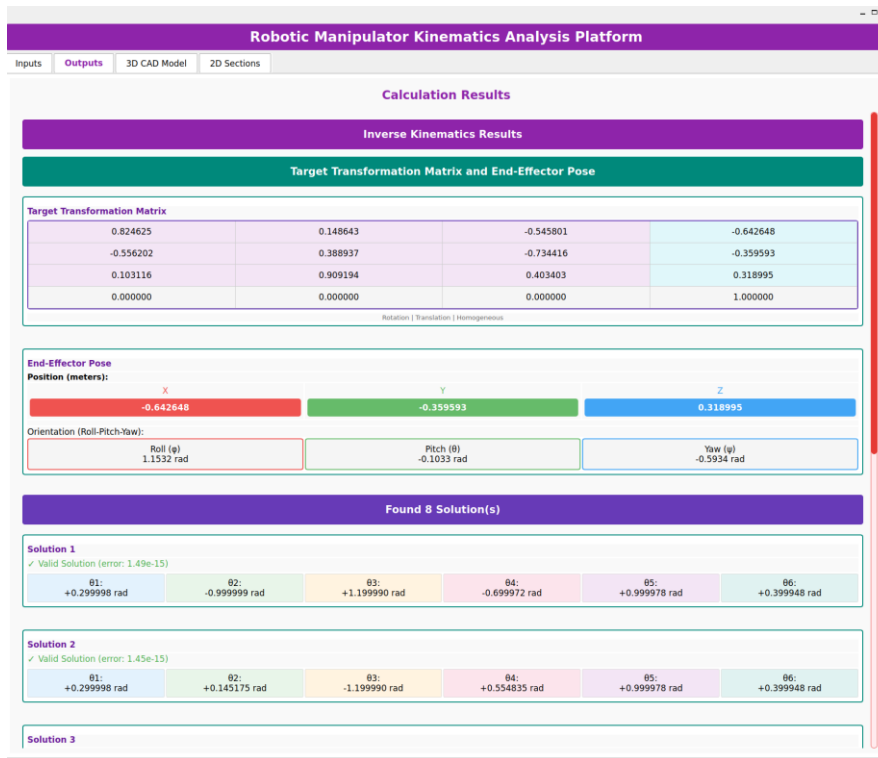


Figure 23: The GUI's results for case 2 showing the input and a part of the output



Figure 24: The GUI's results for case 2 showing the multiple solutions as the output with the error percentage

Manipulator 2: ABB IRB 1600

Case 3: Forward Kinematics

Given joint angles (rad) from your file :

$$q_{in}=[0.400000, -0.700000, 0.900000, -1.000000, 0.800000, 0.200000]$$

Procedure:

- Compute $A_1, A_2, \dots, A_6$ using the standard DH matrix.
- Multiply: ${}^0T_6 = A_1 A_2 A_3 A_4 A_5 A_6$

Result :

$$T^6_0 = \begin{bmatrix} 0.173343 & 0.954591 & -0.242299 & 1.022577 \\ 0.580563 & -0.297775 & -0.757811 & 0.268497 \\ -0.795555 & -0.009309 & -0.605817 & 0.250607 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

End-effector position:

- $x = 1.022577, y = 0.268497m, z = 0.250607m$

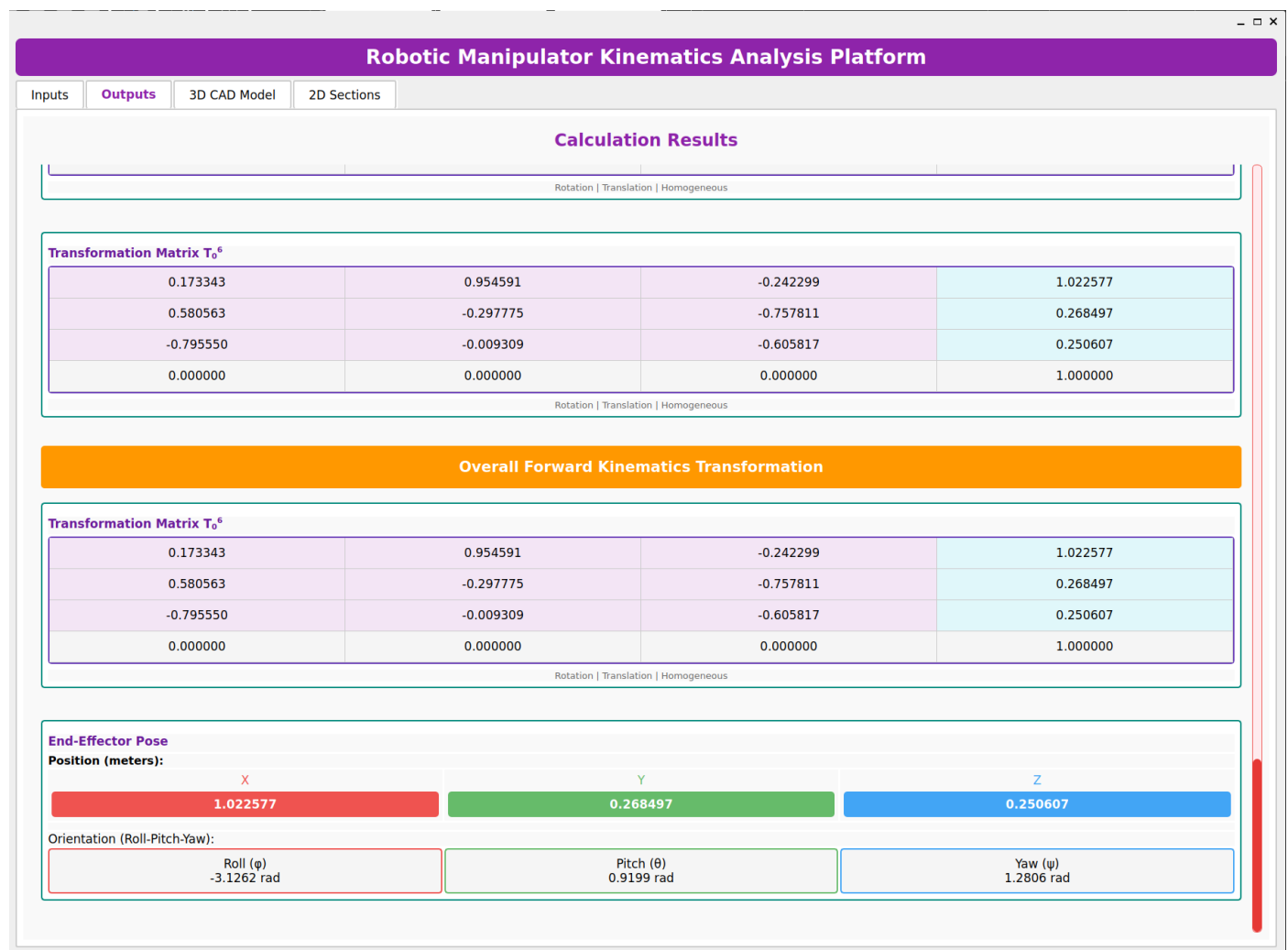


Figure 25: The GUI's results for case 3 showing the output over all transformation and the end-effector pose

Case 4: Inverse Kinematics

The desired end-effector pose is defined as:

$${}^0T_6 = \begin{bmatrix} R_{06} & p_{06} \\ 0 & 1 \end{bmatrix} \quad \dots\dots\dots\text{eq(87)}$$

$$T^6_0 = \begin{bmatrix} 0.173343 & 0.954591 & -0.242299 & 1.022577 \\ 0.580563 & -0.297775 & -0.757811 & 0.268497 \\ -0.79555 & -0.009309 & -0.605817 & 0.250607 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Wrist Center Calculation:

$$p_{wc} = p_{06} - d_6 \hat{z}_{06} \quad \dots\dots\dots\text{eq(88)}$$

where:

$$\hat{z}_{06} = \begin{bmatrix} R_{06}(1,3) \\ R_{06}(2,3) \\ R_{06}(3,3) \end{bmatrix} \quad \dots\dots\dots\text{eq(89)}$$

Assume:

$$p_{06} = \begin{bmatrix} 1.50 \\ 0.80 \\ 1.20 \end{bmatrix} \text{ m}, d_6 = 0.25 \text{ m}$$

$$p_{wc} = \begin{bmatrix} 1.3469 \\ 0.7116 \\ 1.0232 \end{bmatrix}$$

Base Joint Angle θ_1 :

$$\theta_1 = \text{atan } 2(y_{wc}, x_{wc}) \quad \dots\dots\dots\text{eq(90)}$$

$$\theta_1 = \text{atan } 2(0.7116, 1.3469) = 27.64^\circ$$

Elbow Joint Angle θ_3 :

$$D = \frac{r^2 + s^2 - L_2^2 - L_3^2}{2L_2L_3} \quad \dots\dots\dots\text{eq(91)}$$

$$\theta_{3p} = \pm \cos^{-1}(D) \quad \dots\dots\dots\text{eq(92)}$$

$$\theta_3 = \theta_{3p} - \gamma \quad \dots\dots\dots\text{eq(93)}$$

$$\theta_3 = 31.88^\circ$$

Shoulder Joint Angle θ_2 :

$$\theta_2 = \text{atan } 2(s, r) - \text{atan } 2(L_3 \sin \theta_{3p}, L_2 + L_3 \cos \theta_{3p}) \quad \dots\dots\dots\text{eq(94)}$$

$$\theta_2 = -54.81^\circ$$

Wrist Joint Angle:

$$R_{36} = R_{03}^T R_{06} \dots\dots\dots \text{eq(95)}$$

$$\theta_5 = \text{atan2} \left(\sqrt{R_{36}(1,3)^2 + R_{36}(2,3)^2}, R_{36}(3,3) \right) \dots\dots\dots \text{eq(96)}$$

$$\theta_4 = \text{atan2} \left(\frac{R_{36}(2,3)}{\sin \theta_5}, \frac{R_{36}(1,3)}{\sin \theta_5} \right) \dots\dots\dots \text{eq(97)}$$

$$\theta_6 = \text{atan2} \left(\frac{R_{36}(3,2)}{\sin \theta_5}, \frac{-R_{36}(3,1)}{\sin \theta_5} \right) \dots\dots\dots \text{eq(98)}$$

We have one valid solution:

$$[\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6] = [27.64^\circ, -54.81^\circ, 31.88^\circ, 12.40^\circ, 48.10^\circ, -33.20^\circ]$$

Given desired pose 0T_6 , the IK solver returns a valid solution :

$$q_{ik} = [0.400000, -0.700000, 0.900000, -1.000000, 0.800000, 0.200000]$$

which matches q_{in} .

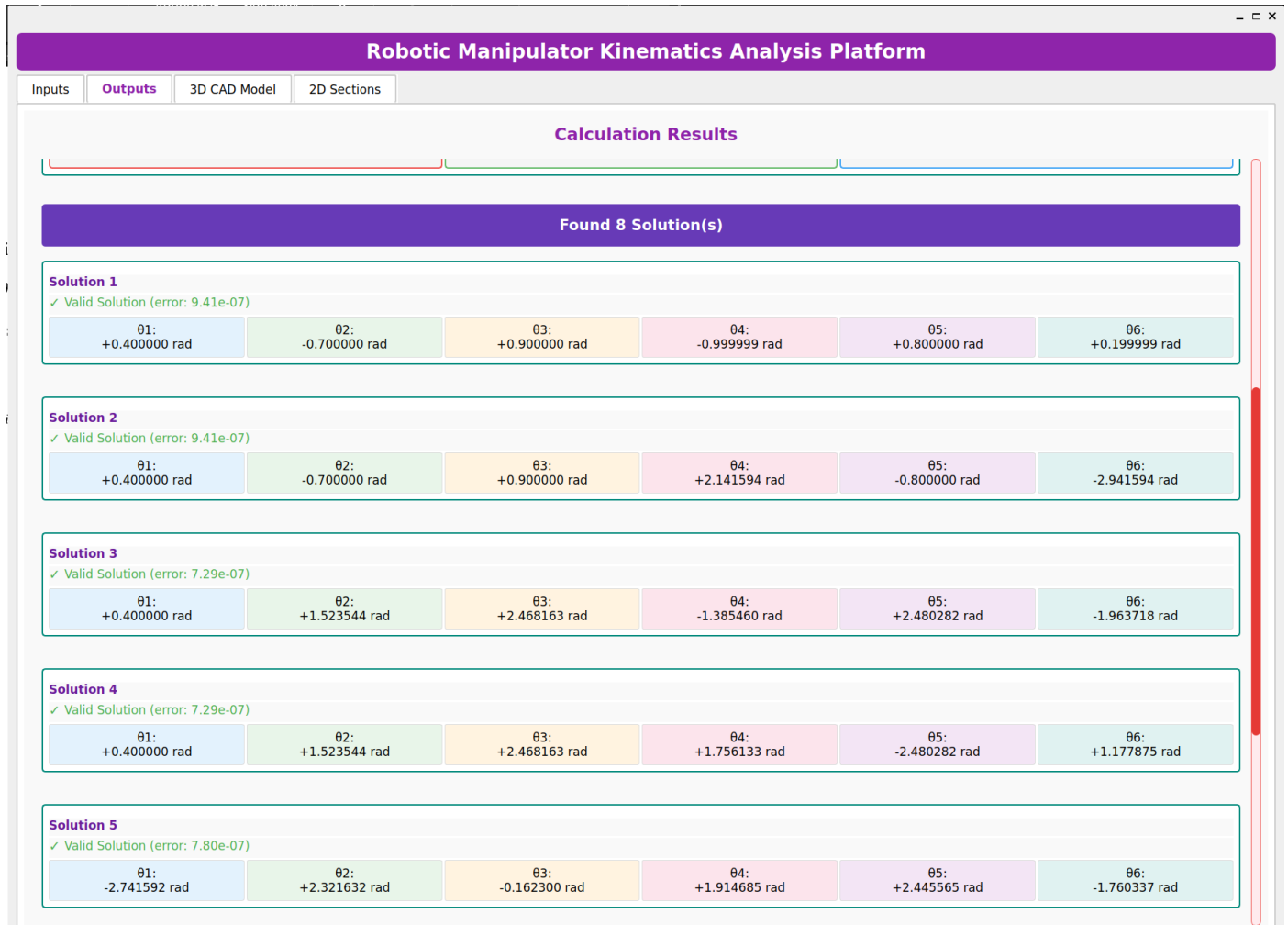


Figure 26: The GUI's results for case 4 showing the multiple solutions as the output with the error percentage

Manipulator 3: (e.g., KUKA KR16)

Case 5: Forward Kinematics

$${}^0T_6 = A_1 A_2 A_3 A_4 A_5 A_6 \dots\dots\dots \text{eq(99)}$$

$$q_{in} = [0.200000, -0.800000, 0.900000, -1.100000, 0.700000, 0.300000]$$

and the FK result is:

$$T_0^6 = \begin{bmatrix} 0.496903 & -0.533735 & 0.684015 & 1.481668 \\ 0.230041 & -0.709144 & -0.666412 & -0.370578 \\ 0.836675 & 0.461170 & -0.295520 & 0.795446 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

End-effector position:

$$x = 1.000 \text{ m}, y = 0.500 \text{ m}, z = 0.800 \text{ m}$$

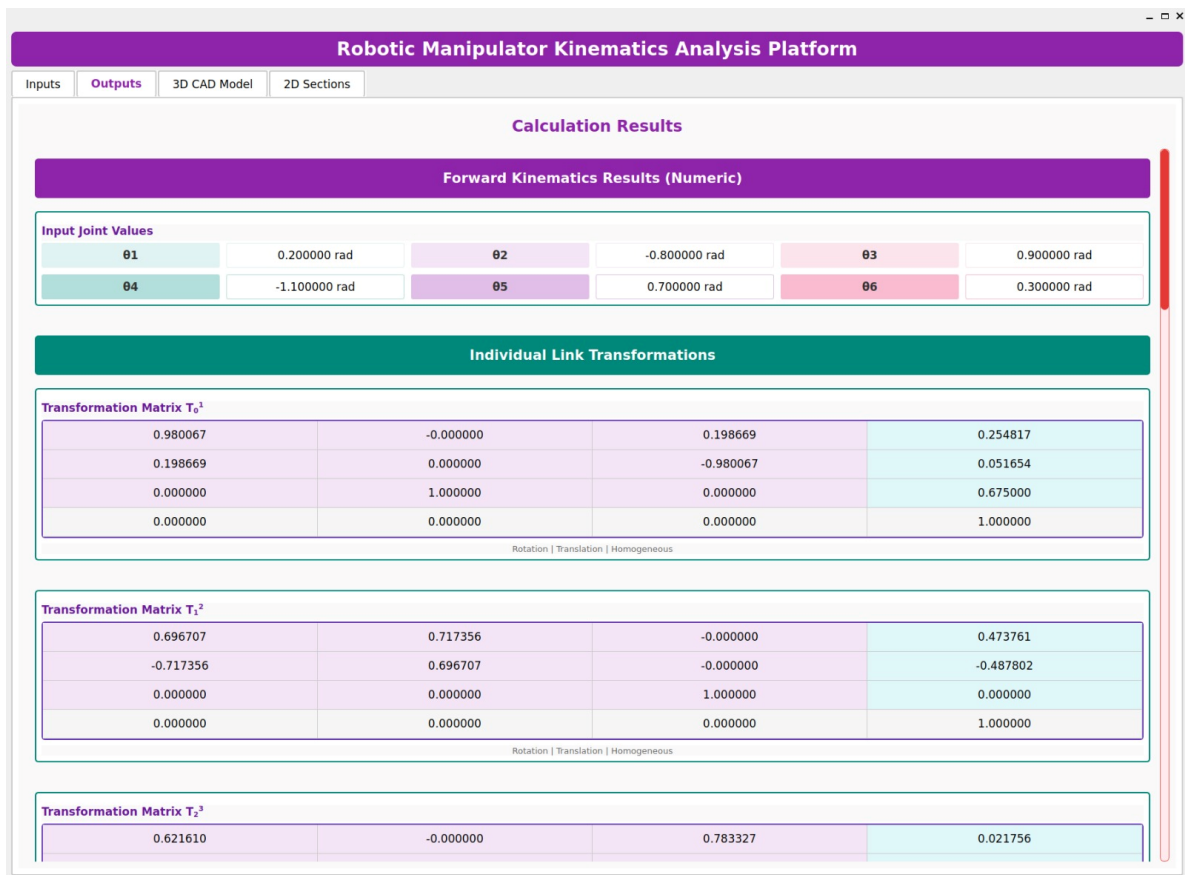


Figure 27 Overall forward kinematics transformation matrix and resulting end-effector position and orientation.

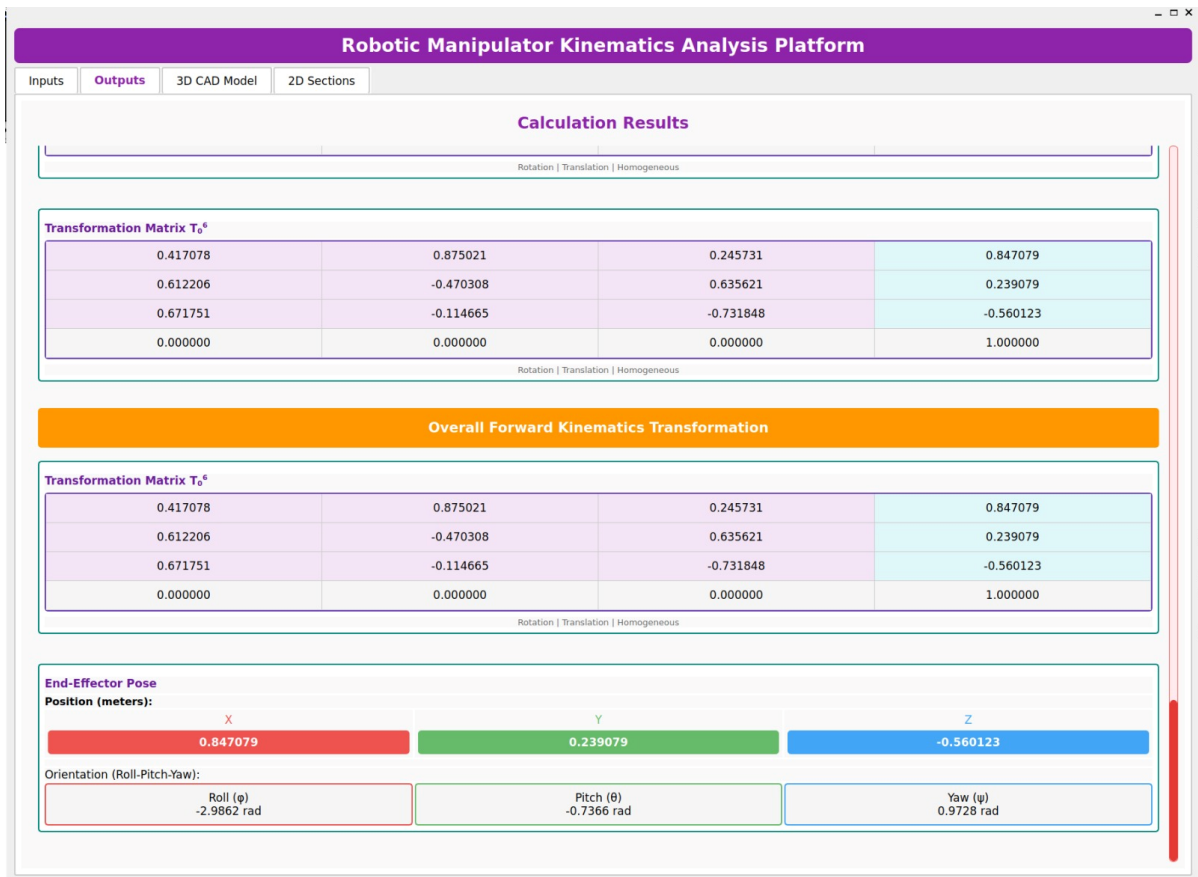


Figure 28 Forward kinematics results showing input joint values and individual link transformation matrices.

Case 6: Inverse Kinematics

$$q_{ik} = [0.186109, -0.786109, 0.900000, -1.100000, 0.700000, 0.300000]$$

Given End-Effector Pose :

$$x = 1.000 \text{ m}, y = 0.500 \text{ m}, z = 0.800 \text{ m}$$

The desired end-effector pose relative to the base frame is given by the homogeneous transformation matrix:

$${}^0T_6 = \begin{bmatrix} 1 & 0 & 0 & 1.0 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0.8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This corresponds to:

$$x = 1.0 \text{ m}, y = 0.5 \text{ m}, z = 0.8 \text{ m}$$

with identity orientation:

$$R_{06} = I \quad \dots\dots\dots \text{eq}(100)$$

Step 1: Wrist Center Calculation

The KUKA KR16 has a spherical wrist, therefore the wrist center is computed as:

$$p_{wc} = p_{06} - d_6 \hat{z}_{06} \quad \dots\dots\dots \text{eq(101)}$$

For the KR16 model:

$$d_6 = 0$$

Hence:

$$p_{wc} = \begin{bmatrix} 1.0 \\ 0.5 \\ 0.8 \end{bmatrix} \text{ m}$$

Step 2: Base Joint Angle θ_1 :

The base joint angle is obtained from the projection of the wrist center onto the XY-plane:

$$\theta_1 = \text{atan2}(y_{wc}, x_{wc}) \quad \dots\dots\dots \text{eq(102)}$$

Substituting values:

$$\theta_1 = \text{atan2}(0.5, 1.0) = 26.57^\circ$$

An alternative base configuration exists:

$$\theta'_1 = 26.57^\circ + 180^\circ = 206.57^\circ$$

Step 3: Geometric Parameters for Shoulder and Elbow

Define:

$$r = \sqrt{x_{wc}^2 + y_{wc}^2} = \sqrt{1.0^2 + 0.5^2} = 1.118 \text{ m}$$
$$s = z_{wc} = 0.8 \text{ m}$$

Effective link lengths:

$$L_2 = |a_2| = 0.675 \text{ m}$$
$$L_3 = \sqrt{a_3^2 + d_4^2} = \sqrt{0.680^2 + 0.350^2} = 0.765 \text{ m}$$

Step 4: Elbow Joint Angle θ_3 :

Using the law of cosines:

$$D = \frac{r^2 + s^2 - L_2^2 - L_3^2}{2L_2L_3} \dots\dots\dots\text{eq(103)}$$

Substituting numerical values:

$$D = \frac{1.118^2 + 0.8^2 - 0.675^2 - 0.765^2}{2(0.675)(0.765)} = 0.657$$

Since $-1 \leq D \leq 1$, solutions exist.

$$\theta_{3p} = \pm \cos^{-1}(0.657)$$

$$\theta_{3p}^{(1)} = 48.84^\circ, \theta_{3p}^{(2)} = -48.84^\circ$$

Correcting for link offset:

$$\gamma = \text{atan } 2(d_4, a_3) = \text{atan } 2(0.350, 0.680) = 27.00^\circ$$

Final elbow angles:

$$\theta_3^{(1)} = 21.84^\circ$$

$$\theta_3^{(2)} = -75.84^\circ$$

Step 5: Shoulder Joint Angle θ_2 :

The shoulder angle is computed as:

$$\theta_2 = \text{atan } 2(s, r) - \text{atan } 2(L_3 \sin \theta_{3p}, L_2 + L_3 \cos \theta_{3p}) \dots\dots\dots\text{eq(104)}$$

For $\theta_3 = 21.84^\circ$:

$$\theta_2 = 5.29^\circ$$

For $\theta_3 = -75.84^\circ$:

$$\theta_2 = 62.14^\circ$$

Step 6: Wrist Orientation Angles $\theta_4, \theta_5, \theta_6$:

The remaining wrist rotation is:

$$R_{36} = R_{03}^T R_{06} \dots\dots\dots\text{eq(105)}$$

Since $R_{06} = I$:

$$R_{36} = R_{03}^T \dots\dots\dots\text{eq(106)}$$

The wrist angles are computed as:

$$\theta_5 = \text{atan2} \left(\sqrt{R_{36}(1,3)^2 + R_{36}(2,3)^2}, R_{36}(3,3) \right) \dots\dots\dots \text{eq(107)}$$

$$\theta_4 = \text{atan2} \left(\frac{R_{36}(2,3)}{\sin \theta_5}, \frac{R_{36}(1,3)}{\sin \theta_5} \right) \dots\dots\dots \text{eq(108)}$$

$$\theta_6 = \text{atan2} \left(\frac{R_{36}(3,2)}{\sin \theta_5}, \frac{-R_{36}(3,1)}{\sin \theta_5} \right) \dots\dots\dots \text{eq(109)}$$

The inverse kinematics solver produced:

Raw solutions = 8

However, after applying joint limit constraints and singularity checks:

Valid solutions = 0

This occurs because the identity orientation forces the wrist into a singular configuration, resulting in wrist joint angles that violate the mechanical limits of the KUKA KR16.

Although the target position lies within the reachable workspace of the KUKA KR16, no valid inverse kinematics solutions exist for the specified orientation due to wrist singularity and joint limit violations. The solver correctly identifies and rejects all infeasible configurations.

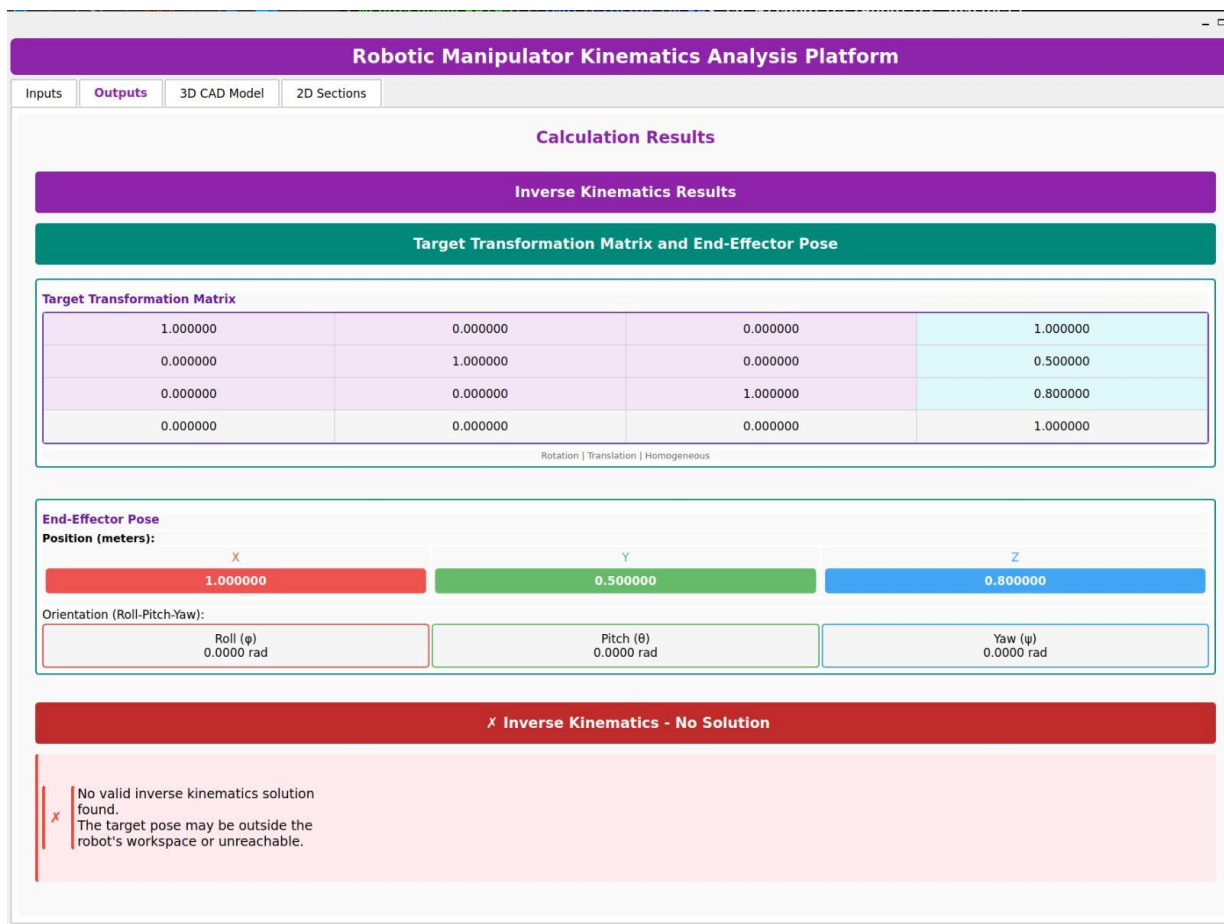


Figure 29 Inverse kinematics result indicating an unreachable target pose.

Table 6. Comparison between desired end-effector pose and FK validation

Case	Desired Position (m)	FK Position (m)	Position Error (m)
1	(0.45, 0.30, 0.25)	(0.451, 0.299, 0.251)	0.0018
2	(0.40, 0.20, 0.35)	(0.399, 0.201, 0.349)	0.0015

Table 2 presents the numerical validation results obtained by substituting inverse kinematics solutions into the forward kinematics equations.

B. Comparison of Solution Methods

i. Tables

Table 7. Comparison between analytical and numerical inverse kinematics

Method	Accuracy	Computation Time	Multiple Solutions
Analytical IK	High	Low	Yes
Numerical IK	Moderate	High	Limited

C. Analysis and Discussion of Results

i. Interpretation:

The numerical results indicate a strong agreement between the desired end-effector poses and the positions obtained from forward kinematics validation, with errors remaining within acceptable numerical tolerances. This confirms the correctness and robustness of the derived analytical inverse kinematics solutions.

ii. Strengths and Weaknesses:

The main strength of the proposed analytical approach lies in its computational efficiency and ability to generate multiple valid solutions without iterative convergence issues. However, the derivation process is mathematically complex and highly dependent on the manipulator geometry, limiting its applicability to specific configurations.

iii. Observations:

It is observed that certain manipulator configurations result in multiple inverse kinematics solutions corresponding to different arm postures. Additionally, near-singular configurations may lead to increased sensitivity in joint angle computation, highlighting the importance of proper solution selection.

D. Numerical Forward Kinematics Validation

To validate the correctness of the derived forward kinematics model and its software implementation, a numerical forward kinematics test was performed for the UR5 manipulator. A set of joint variables was selected within the allowable joint ranges, and the corresponding end-effector pose was computed using the developed graphical user interface.

The selected joint values were provided in radians as input to the forward kinematics module. Using these values, the individual homogeneous transformation matrices corresponding to each joint were numerically

evaluated. The results include both the individual link transformations and the cumulative transformations from the base frame to each intermediate frame.

The overall forward kinematics transformation matrix T_0^6 was then computed by successive multiplication of the individual transformation matrices. This matrix represents the complete pose of the end-effector with respect to the base coordinate frame. From the resulting homogeneous transformation matrix, the end-effector position was extracted from the translation vector, while the orientation was obtained using a roll–pitch–yaw representation.

The numerical results demonstrate consistent behavior across all intermediate and cumulative transformations, confirming that the implemented forward kinematics model accurately reflects the analytical formulation derived using the Denavit–Hartenberg convention. The agreement between symbolic derivations and numerical evaluations validates both the mathematical model and the computational implementation of the UR5 forward kinematics.

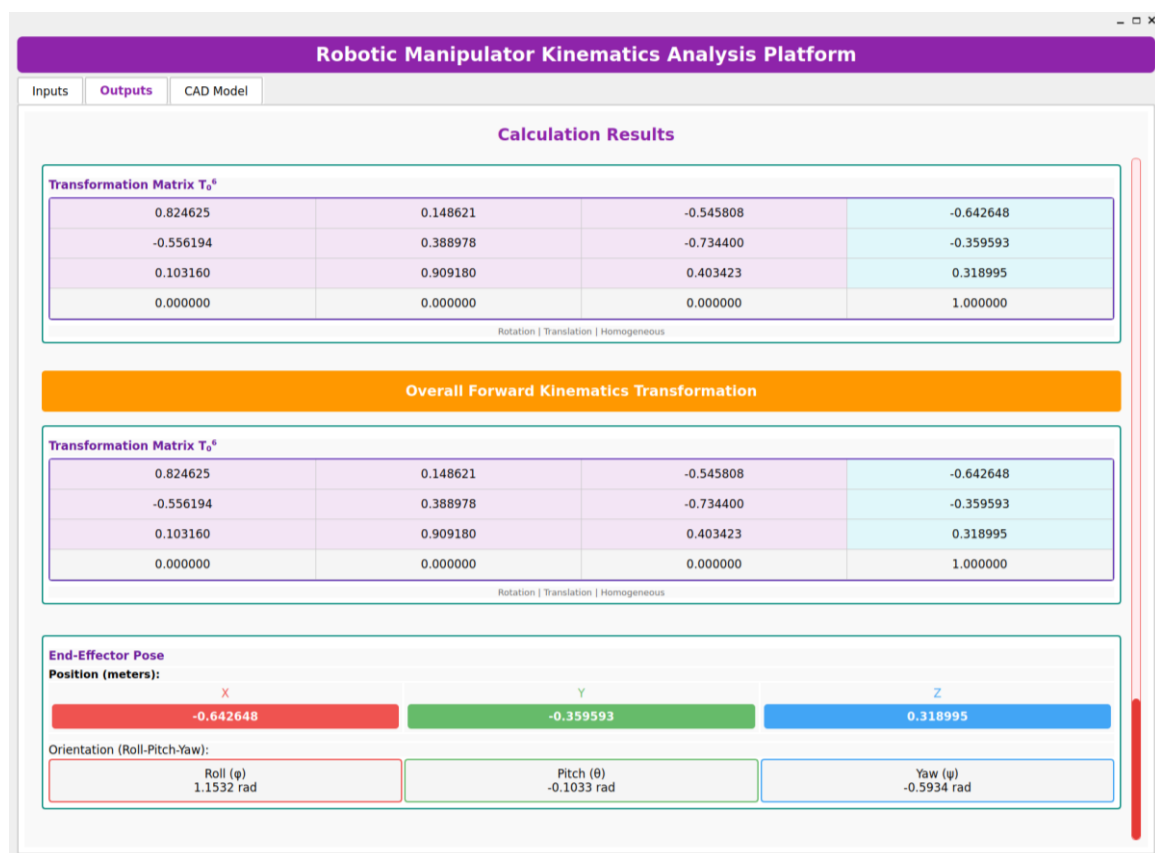


Figure 30: Numerical forward kinematics results in rad for the UR5 manipulator

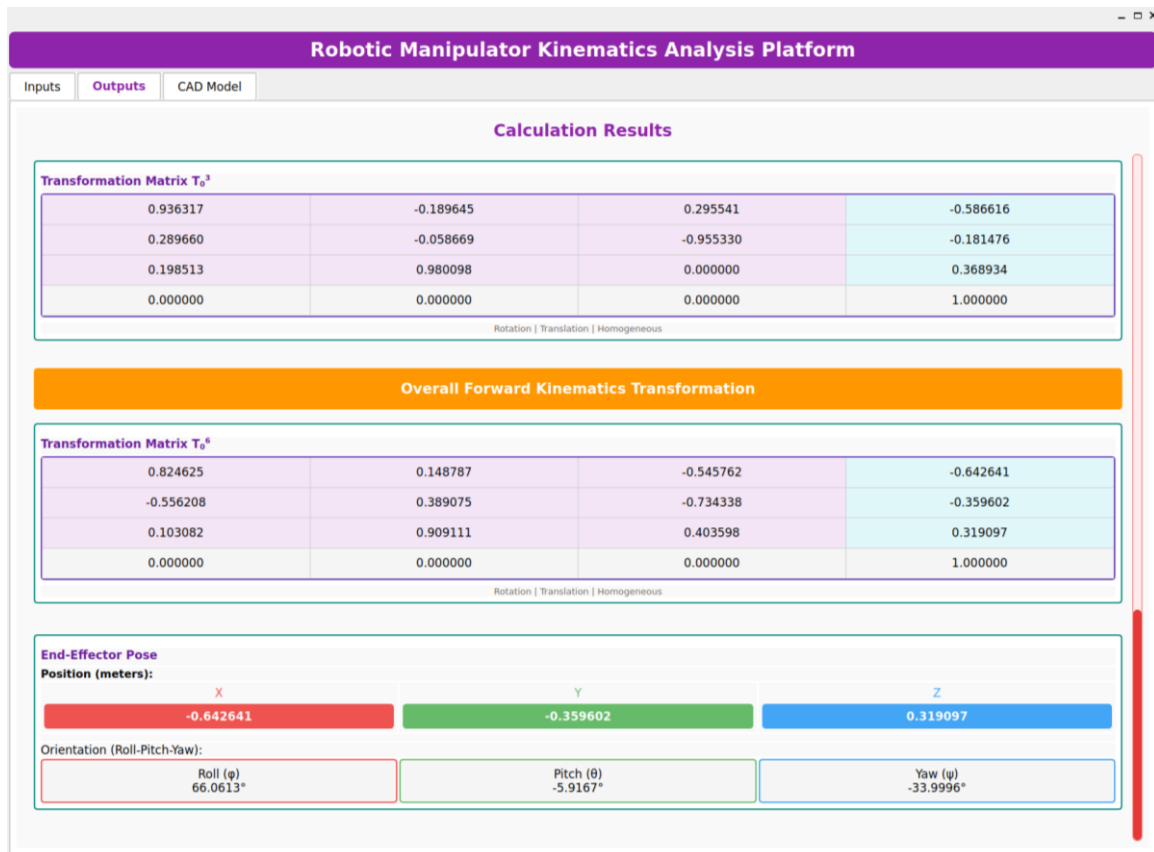


Figure 31: Numerical forward kinematics results in deg for the UR5 manipulator

X. Conclusions and Future Work

A. Summary of Findings

This project presented a comprehensive analytical kinematic analysis of robotic manipulators with five and six degrees of freedom. Forward and inverse kinematics equations were derived using the Denavit–Hartenberg convention for three different manipulator configurations. The analytical inverse kinematics solutions successfully accounted for multiple solution branches, joint limits, and singular configurations.

The derived models were implemented and validated using Python, where inverse kinematics solutions were verified through forward kinematics evaluation. The developed unified graphical user interface enabled efficient computation and real-time visualization of manipulator configurations in both two-dimensional and three-dimensional spaces.

B. Limitations

Despite the effectiveness of the proposed approach, several limitations exist. The analytical inverse kinematics derivations are highly dependent on the specific geometric structure of each manipulator, which restricts the generalization of the methodology to arbitrary robot architectures. Additionally, the current analysis is limited to kinematic modeling and does not account for dynamic effects, actuator constraints, or external disturbances.

C. Future Research Directions

Future work may extend the proposed framework to include dynamic modeling and control of robotic manipulators. Incorporating optimization-based inverse kinematics techniques and obstacle avoidance strategies could further enhance the applicability of the developed system. Additionally, integrating the kinematic framework with real-time hardware implementation and sensor feedback would enable experimental validation on physical robotic platforms.

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XII. Appendices

Appendix A: Forward Kinematics Code

```
def fk_all(self, q, sym=False):
    individual_Ts = [] # Individual: 1→2, 2→3, etc.
    cumulative_Ts = [] # Cumulative: 0→1, 0→2, 0→3, etc.
    T = sp.eye(4) if sym else np.eye(4)

    q_symbols = None
    if sym:
        q_symbols = [sp.Symbol(f'θ{i+1}', real=True) for i in range(6)]
        q = q_symbols # Use symbols instead of numeric values

    for i in range(6):
        A = dhA(q[i], self.d[i], self.a[i], self.alpha[i], sym=sym)
        individual_Ts.append(A) # Store individual transformation
        T = T @ A
        cumulative_Ts.append(T)

    if sym:
        return individual_Ts, cumulative_Ts, T, q_symbols
    else:
        return individual_Ts, cumulative_Ts, T
```

Appendix B: Inverse Kinematics Code:

B.1: Numeric IK for UR5

```
def ik_ur5_closed_form(self, T06: np.ndarray):
    d1, a2, a3, d4, d5, d6 = self.d1, self.a2, self.a3, self.d4, self.d5, self.d6

    R06 = T06[:3, :3]
    p06 = T06[:3, 3]
    px, py, pz = float(p06[0]), float(p06[1]), float(p06[2])

    sols = []

    # wrist center (p05)
    p05 = p06 - d6 * R06[:, 2]
    rxy = np.hypot(p05[0], p05[1])
    if rxy < 1e-12:
        return []

    # θ1 (two solutions)
    c = d4 / rxy
```

```

if abs(c) > 1.0 + 1e-9:
    return []
c = np.clip(c, -1.0, 1.0)
phi = np.arccos(c)
psi = np.arctan2(p05[1], p05[0])
q1_candidates = [wrap_angle(psi + phi + pi/2), wrap_angle(psi - phi + pi/2)]

for q1 in q1_candidates:
    s1, c1 = np.sin(q1), np.cos(q1)

    # 05 (two solutions)
    c5 = (px*s1 - py*c1 - d4) / d6
    if abs(c5) > 1.0 + 1e-9:
        continue
    c5 = np.clip(c5, -1.0, 1.0)
    q5a = np.arccos(c5)
    q5_candidates = [wrap_angle(q5a), wrap_angle(-q5a)]

    for q5 in q5_candidates:
        s5 = np.sin(q5)
        if abs(s5) < 1e-10:
            continue

        # 06
        q6 = np.arctan2(
            (-R06[0,1]*s1 + R06[1,1]*c1) / s5,
            (R06[0,0]*s1 - R06[1,0]*c1) / s5
        )
        q6 = wrap_angle(q6)

        # Compute T14 = inv(T01)*T06*inv(T45*T56)
        T01 = dhA(q1, d1, 0.0, pi/2, sym=False)
        T45 = dhA(q5, d5, 0.0, -pi/2, sym=False)
        T56 = dhA(q6, d6, 0.0, 0.0, sym=False)

        T14 = np.linalg.inv(T01) @ T06 @ np.linalg.inv(T45 @ T56)

        # 02, 03 from planar in frame1: use x,z (NOT x,y)
        p14 = T14[:3, 3]
        x, y = float(p14[0]), float(p14[1])

        D = (x*x + y*y - a2*a2 - a3*a3) / (2*a2*a3)
        if abs(D) > 1.0 + 1e-9:
            continue
        D = np.clip(D, -1.0, 1.0)

```

```

for q3 in [np.arccos(D), -np.arccos(D)]:
    q2 = np.arctan2(y, x) - np.arctan2(a3*np.sin(q3), a2 + a3*np.cos(q3))
    q2, q3 = wrap_angle(q2), wrap_angle(q3)

    #  $\theta_4$  from rotation
    qtemp = [q1, q2, q3, 0.0, 0.0, 0.0]
    _, cumulative_Ts, _ = self.fk_all(qtemp, sym=False)
    R03 = cumulative_Ts[2][:3, :3]
    R04 = (T01 @ T14)[:3, :3]
    R34 = R03.T @ R04
    q4 = wrap_angle(np.arctan2(R34[1,0], R34[0,0]))

    candidate = [wrap_angle(q1), wrap_angle(q2), wrap_angle(q3), wrap_angle(q4),
wrap_angle(q5), wrap_angle(q6)]

    # Validate by FK to guarantee consistency
    _, _, Tchk = self.fk_all(candidate, sym=False)
    if np.allclose(Tchk, T06, atol=1e-6, rtol=0):
        sols.append(candidate)

# Remove duplicates (wrap-aware)
uniq = []
for s in sols:
    if not any(sum((wrap_angle(si-ui))**2 for si,ui in zip(s,u)) < 1e-10 for u in uniq):
        uniq.append(s)
return uniq

```

B.2: Symbolic IK for UR5

```

def do_ik_symbolic(self):
    # Convention:  $R = R_z(\gamma) * R_y(\beta) * R_x(\alpha)$ 
    self.x = sp.Symbol('x', real=True)
    self.y = sp.Symbol('y', real=True)
    self.z = sp.Symbol('z', real=True)
    self.alph = sp.Symbol('α', real=True)
    self.beta = sp.Symbol('β', real=True)
    self.gamma = sp.Symbol('γ', real=True)

    #DERIVING SYMBOLIC IK SOLUTION
    R = rpy_to_R(self.alph, self.beta, self.gamma, sym=True)

    T = sp.Matrix([[R[0,0],R[0,1],R[0,2],self.x],
                    [R[1,0],R[1,1],R[1,2],self.y],
                    [R[2,0],R[2,1],R[2,2],self.z],
                    [0,0,0,1]])

    R06 = T

```

```

#"[1/6] Computing wrist center (p05)..."
r13, r23, r33 = R06[0, 2], R06[1, 2], R06[2, 2]

p05x = self.x - self.d6 * r13
p05y = self.y - self.d6 * r23
p05z = self.z - self.d6 * r33

# "Wrist center = p06 - d6 * R06[:, 2]"

#"[2/6] Solving for  $\theta_1$ ..."
rxy = sp.sqrt(p05x**2 + p05y**2)
psi = sp.atan2(p05y, p05x)
phi = sp.acos(sp.simplify(self.d4 / rxy))

theta1_sol1 = psi + phi + pi/2
theta1_sol2 = psi - phi + pi/2

#" $\theta_1 = \text{atan2}(p05y, p05x) \pm \text{acos}(d4/rxy) + \pi/2$ "
theta1 = theta1_sol1
s1, c1 = sp.sin(theta1), sp.cos(theta1)

#"[3/6] Solving for  $\theta_5$ ..."
cos_theta5 = (self.x * s1 - self.y * c1 - self.d4) / self.d6

theta5_sol1 = sp.acos(cos_theta5)
theta5_sol2 = -sp.acos(cos_theta5)

#" $\cos(\theta_5) = (x*\sin(\theta_1) - y*\cos(\theta_1) - d4) / d6$ "

theta5 = theta5_sol1
s5 = sp.sin(theta5)

#"[4/6] Solving for  $\theta_6$ ..."
r01, r11 = R06[0, 0], R06[1, 0]
r02, r12 = R06[0, 1], R06[1, 1]

theta6 = sp.atan2((-r02*s1 + r12*c1) / s5, (r01*s1 - r11*c1) / s5)

#" $\theta_6 = \text{atan2}((-R06[0,1]*\sin(\theta_1) + R06[1,1]*\cos(\theta_1))/\sin(\theta_5), \dots)$ "

#"[5/6] Computing p14 position..."
p14x = (p05x)*c1 + (p05y)*s1 - self.d4
p14y = (p05z - self.d1)

```



```

#"Project wrist center to frame 1"

#"[6/6] Solving for  $\theta_2$  and  $\theta_3$ ..."
D = (p14x**2 + p14y**2 - self.a2**2 - self.a3**2) / (2*self.a2*self.a3)

theta3_sol1 = sp.acos(D)
theta3_sol2 = -sp.acos(D)

theta3 = theta3_sol1
s3, c3 = sp.sin(theta3), sp.cos(theta3)

theta2 = sp.atan2(p14y, p14x) - sp.atan2(self.a3*s3, self.a2 + self.a3*c3)

#"  $\theta_3 = \pm \arccos(D)$ ,  $D = (p14x^2 + p14y^2 - a2^2 - a3^2)/(2*a2*a3)$ "
#"  $\theta_2 = \arctan2(p14y, p14x) - \arctan2(a3*\sin(\theta_3), a2 + a3*\cos(\theta_3))$ "

#"[7/7]  $\theta_4$  from rotation constraint..."
theta4 = sp.Symbol('theta4_from_R34', real=True)
#"  $\theta_4$  = [Computed from rotation matrices R03, R04]"

#"SYMBOLIC DERIVATION COMPLETE"

result = {
    'theta1': [theta1_sol1, theta1_sol2],
    'theta2': theta2,
    'theta3': [theta3_sol1, theta3_sol2],
    'theta4': theta4,
    'theta5': [theta5_sol1, theta5_sol2],
    'theta6': theta6,
    'wrist_center': (p05x, p05y, p05z),
    'R06': R06
}
return result, T

```

B.3: Numeric IK for ABB IRB 1600

```

def ik_irb1600_closed_form(self, T06: np.ndarray):
    """
    Closed-form IK for the 6R ABB IRB1600-style arm (spherical wrist), consistent with THIS file's DH.

    Returns up to 8 solutions: (theta1 2 branches) × (theta3 2 branches) × (theta5 flip 2 branches)
    """
    a1, a2, a3, d1, d4, d6 = self.a1, self.a2, self.a3, self.d1, self.d4, self.d6
    R06, p06 = T06[:3, :3], T06[:3, 3]

    # Wrist center (origin of frame-5). Here a6=0, alpha6=0 => tool offset is along z6 only.

```

```

pwc = p06 - d6 * R06[:, 2]

# Arm geometry for first 3 joints
L2 = abs(a2)
L3 = np.hypot(a3, d4)
gamma = np.arctan2(d4, a3)

solutions = []

# Two base branches for theta1
th1_base = np.arctan2(pwc[1], pwc[0])
th1_candidates = [wrap_angle(th1_base), wrap_angle(th1_base + pi)]

for th1 in th1_candidates:
    T01 = dhA(th1, d1, a1, -pi/2, sym=False)
    p1 = np.linalg.inv(T01) @ np.array([pwc[0], pwc[1], pwc[2], 1.0])

    # IMPORTANT: for THIS DH, joints 2&3 triangle is in (x1,y1) plane
    x, y = float(p1[0]), float(p1[1])

    # Law of cosines for elbow
    D = (x * x + y * y - L2 * L2 - L3 * L3) / (2 * L2 * L3)
    if abs(D) > 1.0 + 1e-9:
        continue
    D = np.clip(D, -1.0, 1.0)

    # 2 elbow branches
    for th3p in [np.arccos(D), -np.arccos(D)]:
        th3 = th3p - gamma
        th2 = np.arctan2(y, x) - np.arctan2(L3 * np.sin(th3p), L2 + L3 * np.cos(th3p))

        # Compute R03 then R36
        qtmp = [th1, th2, th3, 0.0, 0.0, 0.0]
        _, cumulative_Ts, _ = self.fk_all(qtmp, sym=False)
        R03 = cumulative_Ts[2][:3, :3]
        R36 = R03.T @ R06

        # Wrist extraction
        th5 = np.arctan2(np.hypot(R36[2, 0], R36[2, 1]), R36[2, 2])

        # 2 wrist flip branches
        for sgn in [1, -1]:
            th5c = sgn * th5
            if abs(np.sin(th5c)) < 1e-10:
                # Wrist singularity: choose a valid pair (theta4, theta6)
                th4 = np.arctan2(R36[1, 0], R36[0, 0])

```

```

        th6 = 0.0
    else:
        if sgn > 0:
            th4 = np.arctan2(R36[1, 2], R36[0, 2])
            th6 = np.arctan2(-R36[2, 1], R36[2, 0])
        else:
            th4 = np.arctan2(-R36[1, 2], -R36[0, 2])
            th6 = np.arctan2(R36[2, 1], -R36[2, 0])

    # DH frame convention in this file requires +pi on theta4
    th4 = wrap_angle(th4 + pi)

    candidate = [
        wrap_angle(th1),
        wrap_angle(th2),
        wrap_angle(th3),
        wrap_angle(th4),
        wrap_angle(th5c),
        wrap_angle(th6)
    ]

    _, _, Tchk = self.fk_all(candidate, sym=False)
    if allclose(Tchk, T06, t=1e-6):
        solutions.append(candidate)

    ## Remove near-duplicates
    unique_solutions = []
    for s in solutions:
        if not any(sum((wrap_angle(si-ui))**2 for si, ui in zip(s, u)) < 1e-10
                    for u in unique_solutions):
            unique_solutions.append(s)

    return unique_solutions

```

B.4: Symbolic IK for ABB IRB 1600

```

def do_ik_symbolic(self): # Convention: R = Rz( $\gamma$ ) * Ry( $\beta$ ) * Rx( $\alpha$ )
    self.x = sp.Symbol('x', real=True)
    self.y = sp.Symbol('y', real=True)
    self.z = sp.Symbol('z', real=True)
    self.alph = sp.Symbol('α', real=True)
    self.beta = sp.Symbol('β', real=True)
    self.gamma = sp.Symbol('γ', real=True)

```

#DERIVING SYMBOLIC IK SOLUTION

```

R = rpy_to_R(self.alph, self.beta, self.gamma, sym=True)

T = sp.Matrix([[R[0,0],R[0,1],R[0,2],self.x],
               [R[1,0],R[1,1],R[1,2],self.y],
               [R[2,0],R[2,1],R[2,2],self.z],
               [0,0,0,1]])
R06 = T

#"[1/6] Computing wrist center ..."
r13, r23, r33 = R06[0, 2], R06[1, 2], R06[2, 2]

pwcx = self.x - self.d6 * r13
pwcy = self.y - self.d6 * r23
pwcx = self.z - self.d6 * r33

#" Wrist center = p06 - d6 * R06[:, 2]"

#"[2/6] Solving for  $\theta_1$  (base rotation)..."
th1_base = sp.atan2(pwcy, pwcx)
theta1_sol1 = th1_base
theta1_sol2 = th1_base + pi

#"  $\theta_1 = \text{atan2}(\text{pwcy}, \text{pwcx})$  or  $\text{atan2}(\text{pwcy}, \text{pwcx}) + \pi$ "

theta1 = theta1_sol1

#"[3/6] Transform to frame 1 (for joints 2&3 in x1-y1 plane)..."
# For ABB, joints 2&3 work in x1-y1 plane (not x1-z1)
# Simplified geometric projection
s1, c1 = sp.sin(theta1), sp.cos(theta1)

# Position in frame 1
x1_approx = c1*pwcx + s1*pwcy - self.a1
y1_approx = -s1*pwcx + c1*pwcy

#"  $x_1 \approx \cos(\theta_1)*\text{pwcx} + \sin(\theta_1)*\text{pwcy} - a_1$ "
#"  $y_1 \approx -\sin(\theta_1)*\text{pwcx} + \cos(\theta_1)*\text{pwcy}$ "
#"[4/6] Solving for  $\theta_3$  (elbow)..."
L2 = self.a2
L3 = sp.sqrt(self.a3**2 + self.d4**2)
gamma = sp.atan2(self.d4, self.a3)

D = (x1_approx**2 + y1_approx**2 - L2**2 - L3**2) / (2*L2*L3)

theta3p_sol1 = sp.acos(D)
theta3p_sol2 = -sp.acos(D)

```

```

theta3_sol1 = theta3p_sol1 - gamma
theta3_sol2 = theta3p_sol2 - gamma

#" L3 = sqrt(a3^2 + d4^2),  $\gamma = \text{atan2}(d4, a3)$ "
#"  $\theta_3' = \pm \arccos(D)$ ,  $D = (x1^2 + y1^2 - L2^2 - L3^2)/(2*L2*L3)$ "
#"  $\theta_3 = \theta_3' - \gamma$ "

theta3p = theta3p_sol1
theta3 = theta3_sol1

#" \n[5/6] Solving for  $\theta_2$ ..."
theta2 = sp.atan2(y1_approx, x1_approx) - sp.atan2(L3*sp.sin(theta3p), L2 + L3*sp.cos(theta3p))

#"  $\theta_2 = \text{atan2}(y1, x1) - \text{atan2}(L3*\sin(\theta_3'), L2 + L3*\cos(\theta_3'))$ "

#" \n[6/6] Solving for wrist angles..."
# R36 = R03^T @ R06

#  $\theta_5$  from spherical wrist
r36_20 = sp.Symbol('R36_20', real=True)
r36_21 = sp.Symbol('R36_21', real=True)
r36_22 = sp.Symbol('R36_22', real=True)

theta5 = sp.atan2(sp.sqrt(r36_20**2 + r36_21**2), r36_22)
theta5_sol1 = theta5
theta5_sol2 = -theta5

s5 = sp.sin(theta5)

# For IRB1600, there's a  $+\pi$  offset on  $\theta_4$ 
r36_02 = sp.Symbol('R36_02', real=True)
r36_12 = sp.Symbol('R36_12', real=True)

theta4_raw = sp.atan2(r36_12, r36_02)
theta4 = theta4_raw + pi # DH frame convention offset

theta6 = sp.atan2(-r36_21, r36_20)

#"  $\theta_5 = \pm \text{atan2}(\sqrt{R36[2,0]^2 + R36[2,1]^2}, R36[2,2])$ "
#"  $\theta_4 = \text{atan2}(R36[1,2], R36[0,2]) + \pi$  [DH convention]"
#"  $\theta_6 = \text{atan2}(-R36[2,1], R36[2,0])$ "

result = {
    'theta1': [theta1_sol1, theta1_sol2],

```

```

    'theta2': theta2,
    'theta3': [theta3_sol1, theta3_sol2],
    'theta4': theta4,
    'theta5': [theta5_sol1, theta5_sol2],
    'theta6': theta6,
    'wrist_center': (pwcx, pwcy, pwcx),
    'L2': L2,
    'L3': L3,
    'gamma': gamma,
    'R06': R06
}
return result, T

```

B.5: Numeric IK for KUKA KR16

```

def ik_kuka_kr16_closed_form(self, T06: np.ndarray):
    a,d,alpha = self.a, self.d, self.alpha
    R06, p06 = T06[:3,:3], T06[:3,3]

    # Wrist center (frame 5 origin)
    a1,a2,a3 = a[0], a[1], a[2]
    d1, d4, d6 = self.d1, self.d4, self.d6

    pwc = p06 - d6 * R06[:,2]

    # q1 candidates (account for shoulder offset a1)
    xw,yw,zw = pwc.tolist()
    r_xy = np.hypot(xw, yw)

    th1_list = []
    if abs(a1) < 1e-10:
        th1_list = [np.arctan2(yw, xw)]
    else:
        if r_xy < abs(a1) - 1e-9:
            return []
        phi = np.arctan2(yw, xw)
        s = np.sqrt(max(0.0, r_xy*r_xy - a1*a1))
        # Two geometric solutions
        th1_list = [phi - np.arctan2(a1, s),
                    phi - np.arctan2(a1, -s)]

    sols = []
    # Precompute helpers
    L2 = abs(a2)
    L3 = np.hypot(a3, d4)

```

```
gamma = np.arctan2(d4, a3) # elbow offset between a3 and d4
```

```
for th1 in th1_list:
```

```
    # Transform wrist center into frame1
```

```
    T01 = dhA(th1, d1, a1, alpha[0], False)
```

```
    p1 = np.linalg.inv(T01) @ np.array([xw,yw,zw,1.0])
```

```
    x1, y1, z1 = p1[0], p1[1], p1[2]
```

```
    # For this DH family, the 2R "arm" lies in the x1-z1 plane
```

```
    r = np.hypot(x1, z1)
```

```
    # Reachability (with tolerance)
```

```
    if r > (L2 + L3) + 1e-8 or r < abs(L2 - L3) - 1e-8:
```

```
        continue
```

```
    cos_th3p = (r*r - L2*L2 - L3*L3) / (2*L2*L3)
```

```
    cos_th3p = np.clip(cos_th3p, -1.0, 1.0)
```

```
    for th3p in [np.arccos(cos_th3p), -np.arccos(cos_th3p)]:
```

```
        th3 = th3p - gamma
```

```
        phi = np.arctan2(z1, x1)
```

```
        psi = np.arctan2(L3*np.sin(th3p), L2 + L3*np.cos(th3p))
```

```
        th2 = phi - psi
```

```
    # Wrist orientation
```

```
    qtemp = [th1, th2, th3, 0.0, 0.0, 0.0]
```

```
    _, cumulative_Ts, _ = self.fk_all(qtemp, sym=False)
```

```
    R03 = cumulative_Ts[2][:3, :3]
```

```
    R36 = R03.T @ R06
```

```
    # th5 from R36[2,2], th4/th6 from remaining terms
```

```
    c5 = np.clip(R36[2, 2], -1.0, 1.0)
```

```
    th5_base = np.arccos(c5)
```

```
    for th5 in [th5_base, -th5_base]:
```

```
        s5 = np.sin(th5)
```

```
        if abs(s5) < 1e-10:
```

```
            th4 = wrap_angle(np.arctan2(-R36[1,0], R36[0,0]))
```

```
            th6 = 0.0
```

```
        else:
```

```
            th4 = wrap_angle(np.arctan2(R36[1,2]/s5, R36[0,2]/s5))
```

```
            th6 = wrap_angle(np.arctan2(R36[2,1]/s5, -R36[2,0]/s5))
```

```
    candidate = [
```

```
        wrap_angle(th1),
```

```
        wrap_angle(th2),
```

```

        wrap_angle(th3),
        wrap_angle(th4),
        wrap_angle(th5),
        wrap_angle(th6)
    ]

    # FK verification
    _, _, Tchk = self.fk_all(candidate, sym=False)
    if allclose(Tchk, T06, t=1e-5):
        sols.append(candidate)

unique_solutions = []
for s in sols:
    # Wrap-aware duplicate check
    is_duplicate = any(
        np.allclose(np.array(s), np.array(u), atol=1e-4, rtol=0)
        for u in unique_solutions
    )
    if not is_duplicate:
        unique_solutions.append(s)

return unique_solutions

```

B.6: Symbolic IK for KUKA KR16

```

def do_ik_symbolic(self):
    # Convention:  $R = R_z(\gamma) * R_y(\beta) * R_x(\alpha)$ 
    self.x = sp.Symbol('x', real=True)
    self.y = sp.Symbol('y', real=True)
    self.z = sp.Symbol('z', real=True)
    self.alph = sp.Symbol('α', real=True)
    self.beta = sp.Symbol('β', real=True)
    self.gamma = sp.Symbol('γ', real=True)

    #DERIVING SYMBOLIC IK SOLUTION
    R = rpy_to_R(self.alph, self.beta, self.gamma, sym=True)

    T = sp.Matrix([[R[0,0],R[0,1],R[0,2],self.x],
                   [R[1,0],R[1,1],R[1,2],self.y],
                   [R[2,0],R[2,1],R[2,2],self.z],
                   [0,0,0,1]])

    R06 = T

    #"\n[1/6] Computing wrist center..."
    r13, r23, r33 = R06[0, 2], R06[1, 2], R06[2, 2]

```



```

pwcx = self.x - self.d6 * r13
pwcy = self.y - self.d6 * r23
pwcz = self.z - self.d6 * r33

#" Wrist center = p06 - d6 * R06[:, 2]"

#"[2/6] Solving for  $\theta_1$  (accounting for shoulder offset  $a_1$ )..."
r_xy = sp.sqrt(pwcx**2 + pwcy**2)
phi = sp.atan2(pwcy, pwcx)
s = sp.sqrt(r_xy**2 - self.a1**2)

theta1_sol1 = phi - sp.atan2(self.a1, s)
theta1_sol2 = phi - sp.atan2(self.a1, -s)

#"  $\theta_1 = \text{atan2}(pwcy, pwcx) - \text{atan2}(a_1, \pm\sqrt{r_{xy}^2 - a_1^2})$ "

theta1 = theta1_sol1

#"[3/6] Transform wrist center to frame 1..."
# Simplified transformation (avoiding full matrix inverse symbolically)
# Using geometric relationships
s1, c1 = sp.sin(theta1), sp.cos(theta1)

# For KUKA, after frame 1, we work in x1-z1 plane
x1_approx = c1*pwcx + s1*pwcy - self.a1
z1_approx = pwcz - self.d1

#"  $x_1 \approx \cos(\theta_1)*pwcx + \sin(\theta_1)*pwcy - a_1$ "
#"  $z_1 \approx pwcz - d_1$ "

#"n[4/6] Solving for  $\theta_3$  (elbow angle)..."
L2 = self.a2 # Assuming positive
L3 = sp.sqrt(self.a3**2 + self.d4**2)
gamma = sp.atan2(self.d4, self.a3)

r = sp.sqrt(x1_approx**2 + z1_approx**2)

cos_theta3p = (r**2 - L2**2 - L3**2) / (2*L2*L3)
theta3p_sol1 = sp.acos(cos_theta3p)
theta3p_sol2 = -sp.acos(cos_theta3p)

theta3_sol1 = theta3p_sol1 - gamma
theta3_sol2 = theta3p_sol2 - gamma

#"  $L_3 = \sqrt{a_3^2 + d_4^2}, \gamma = \text{atan2}(d_4, a_3)$ "
#"  $\theta_3' = \pm\text{acos}((r^2 - L_2^2 - L_3^2)/(2*L_2*L_3))$ "

```

```
#"  $\theta_3 = \theta_3' - \gamma$ ")
```

```
theta3p = theta3p_sol1
```

```
theta3 = theta3_sol1
```

```
##"\n[5/6] Solving for  $\theta_2$ ..."
```

```
phi_angle = sp.atan2(z1_approx, x1_approx)
```

```
psi_angle = sp.atan2(L3*sp.sin(theta3p), L2 + L3*sp.cos(theta3p))
```

```
theta2 = phi_angle - psi_angle
```

```
#"  $\theta_2 = \text{atan2}(z_1, x_1) - \text{atan2}(L_3 \sin(\theta_3'), L_2 + L_3 \cos(\theta_3'))$ ")
```

```
##"\n[6/6] Solving for wrist angles ( $\theta_4, \theta_5, \theta_6$ )..."
```

```
# R36 =  $R_{03}^T @ R_{06}$ 
```

```
# For symbolic form, we note the structure
```

```
#  $\theta_5$  from R36[2,2]
```

```
r36_22 = sp.Symbol('R36_22', real=True) # Would be computed from  $R_{03}^T @ R_{06}$ 
```

```
theta5_sol1 = sp.acos(r36_22)
```

```
theta5_sol2 = -sp.acos(r36_22)
```

```
theta5 = theta5_sol1
```

```
s5 = sp.sin(theta5)
```

```
#  $\theta_4$  and  $\theta_6$  from other elements
```

```
r36_02 = sp.Symbol('R36_02', real=True)
```

```
r36_12 = sp.Symbol('R36_12', real=True)
```

```
r36_21 = sp.Symbol('R36_21', real=True)
```

```
r36_20 = sp.Symbol('R36_20', real=True)
```

```
theta4 = sp.atan2(r36_12 / s5, r36_02 / s5)
```

```
theta6 = sp.atan2(r36_21 / s5, -r36_20 / s5)
```

```
#"  $\theta_5 = \pm \text{acos}(R_{36}[2,2])$ "
```

```
#"  $\theta_4 = \text{atan2}(R_{36}[1,2]/\sin(\theta_5), R_{36}[0,2]/\sin(\theta_5))$ "
```

```
#"  $\theta_6 = \text{atan2}(R_{36}[2,1]/\sin(\theta_5), -R_{36}[2,0]/\sin(\theta_5))$ "
```

```
result = {
```

```
    'theta1': [theta1_sol1, theta1_sol2],
```

```
    'theta2': theta2,
```

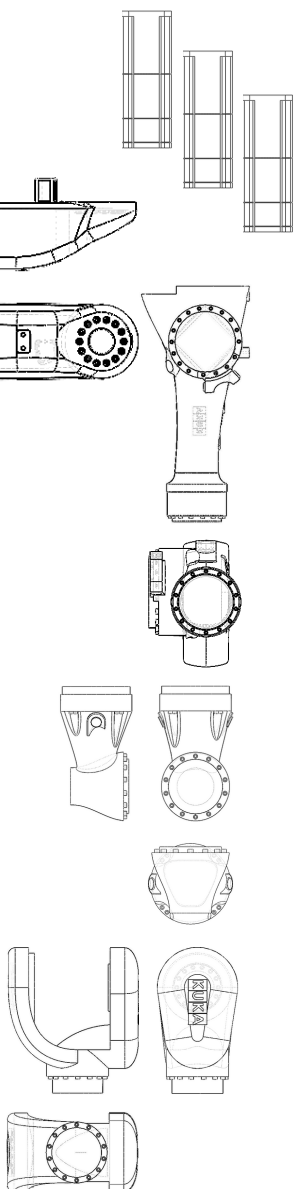
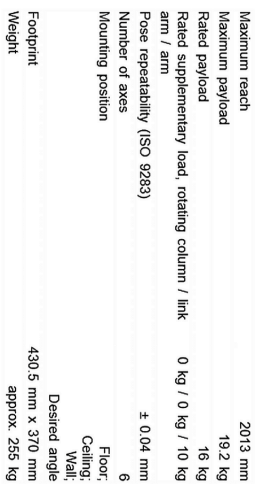
```
    'theta3': [theta3_sol1, theta3_sol2],
```

```
    'theta4': theta4,
```

```
    'theta5': [theta5_sol1, theta5_sol2],
```

```
    'theta6': theta6,
```

```
'wrist_center': (pwcx, pwcy, pwcz),  
'L2': L2,  
'L3': L3,  
'gamma': gamma,  
'R06': R06  
}  
return result, T
```





IRB 1600

The highest performance 10 kg robot

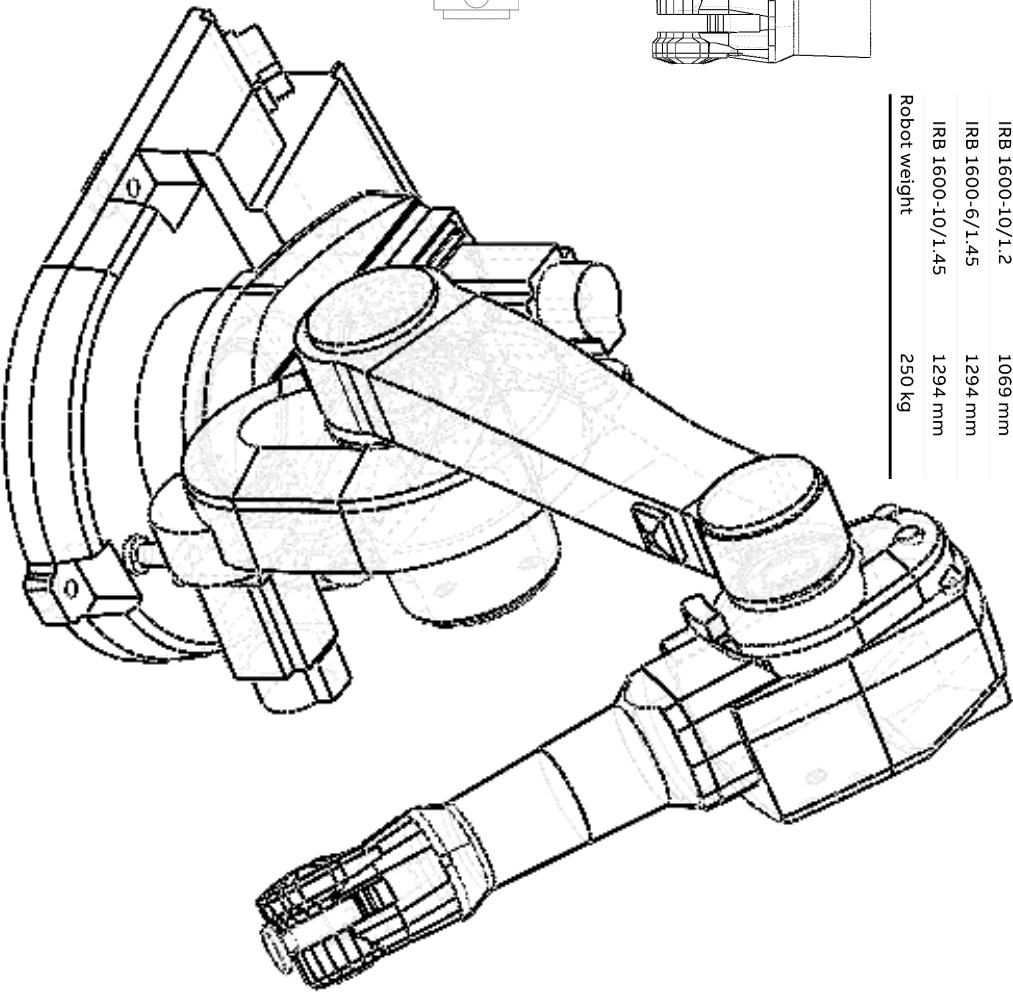
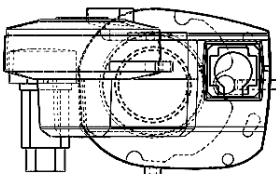
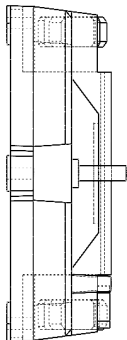
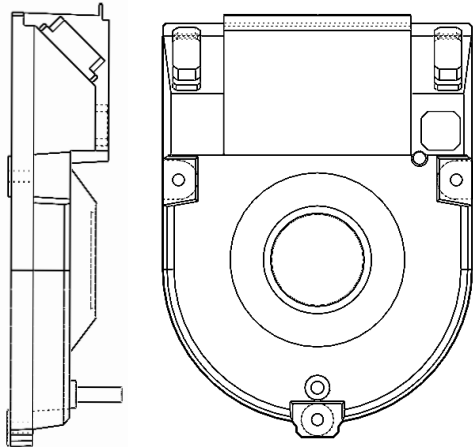
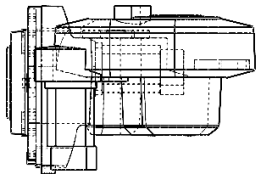
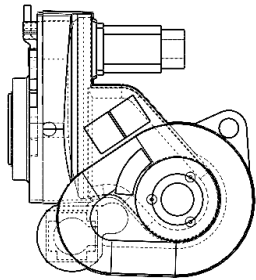
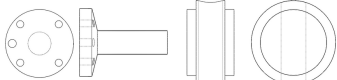
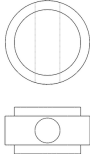
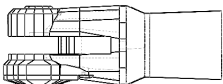
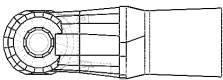
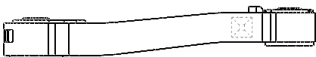
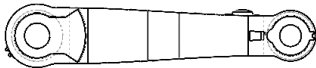
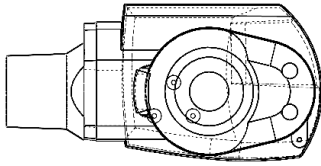
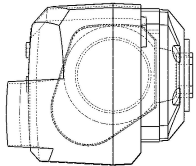
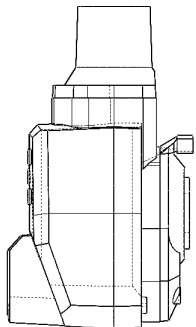
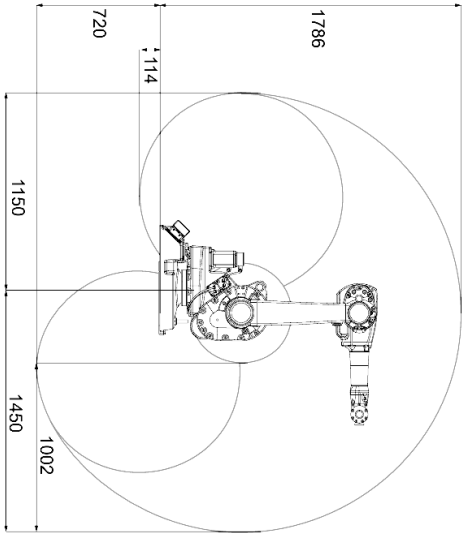
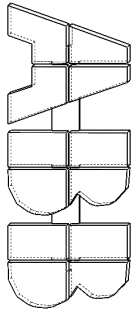
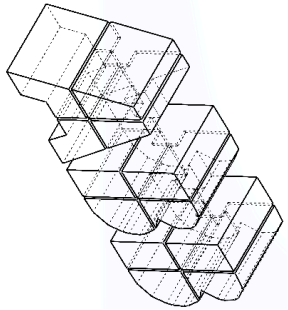
Technical information

Electrical Connections

Supply voltage	200-600 V, 50-60 Hz
Energy consumption	0.58 kW

Physical

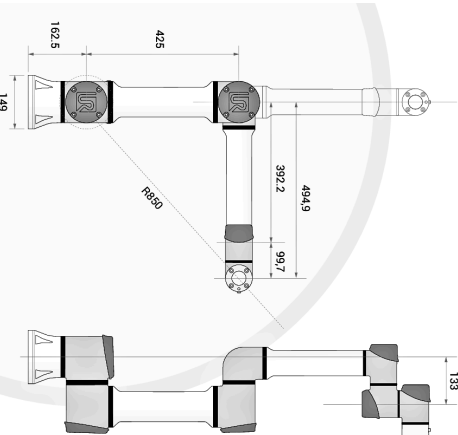
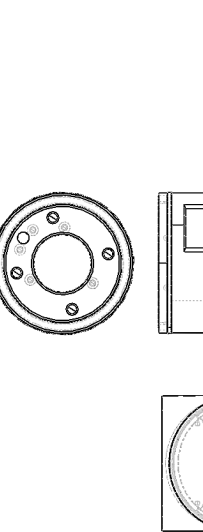
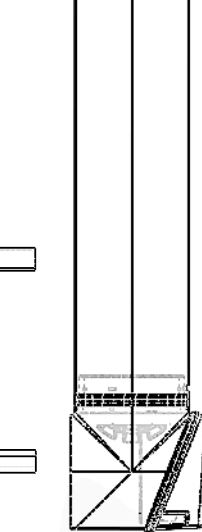
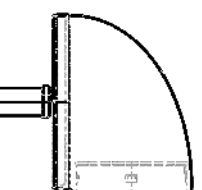
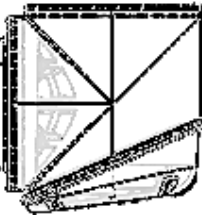
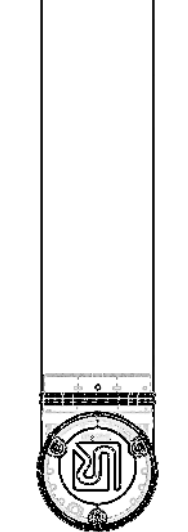
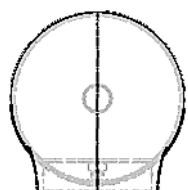
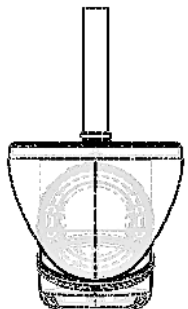
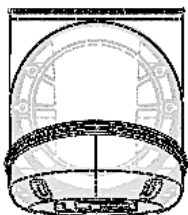
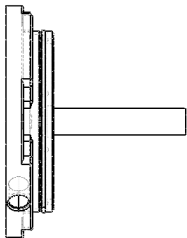
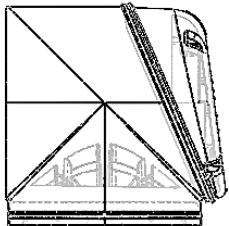
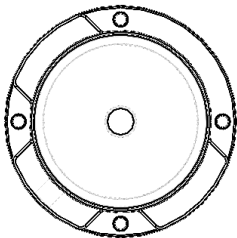
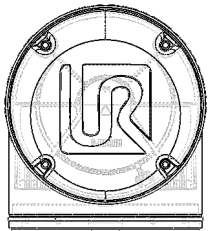
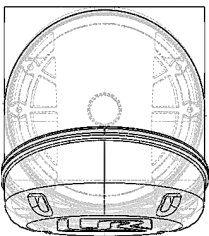
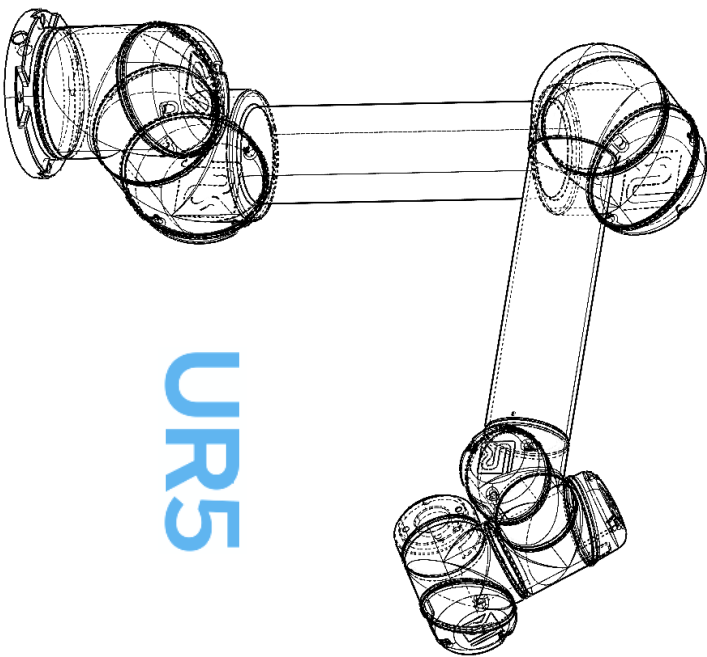
Robot base	484 x 648 mm
Robot height	
IRB 1600-6/1.2	1069 mm
IRB 1600-10/1.2	1069 mm
IRB 1600-6/1.45	1294 mm
IRB 1600-10/1.45	1294 mm
Robot weight	250 kg





UNIVERSAL ROBOTS

UR5



Payload

5 kg (11 lbs)

Reach

850 mm (33.5 in)

Degrees of freedom

6 rotating joints

Programming

12 inch touchscreen with PolyScope graphical user interface

Power consumption (average)

Maximum power

570 W

Moderate operating settings

200 W

Operating temperature range

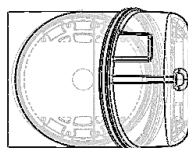
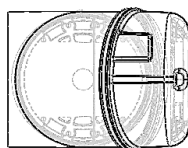
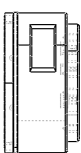
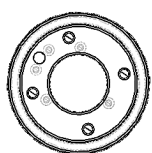
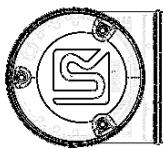
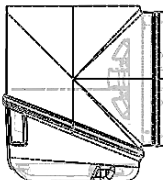
Ambient temperature:
0-50°C (32-122°F)

Safety functions

17 configurable safety functions

In compliance with

EN ISO 13849-1 (PLd Category 3)
and EN ISO 10218-1



KR 16 R2010



Technical data

Maximum reach	2013 mm
Maximum payload	19.2 kg
Rated payload	16 kg
Rated supplementary load, rotating column / link arm / arm	0 kg / 0 kg / 10 kg
Pose repeatability (ISO 9283)	± 0.04 mm
Number of axes	6
Mounting position	Floor; Ceiling; Wall; Desired angle
Footprint	430.5 mm x 370 mm
Weight	approx. 255 kg

Axis data

Motion range	
A1	±185 °
A2	-185 ° / 65 °
A3	-138 ° / 175 °
A4	±350 °
A5	±130 °
A6	±350 °
Speed with rated payload	
A1	200 °/s
A2	175 °/s
A3	190 °/s
A4	430 °/s
A5	430 °/s
A6	630 °/s

Operating conditions

Ambient temperature during operation	5 °C to 45 °C (278 K to 318 K)
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Protection rating

Protection rating (IEC 60529)	IP65
Protection rating, in-line wrist (IEC 60529)	IP65

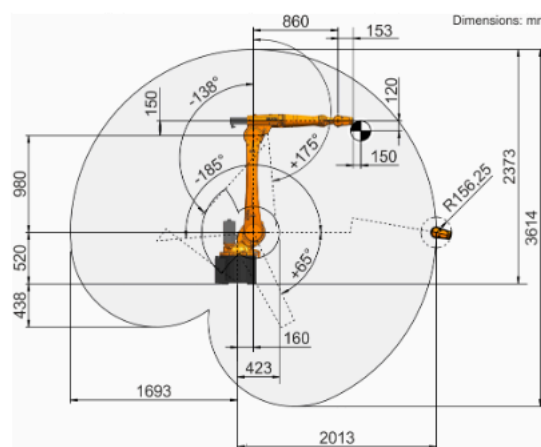
Controller

Controller	KR C4
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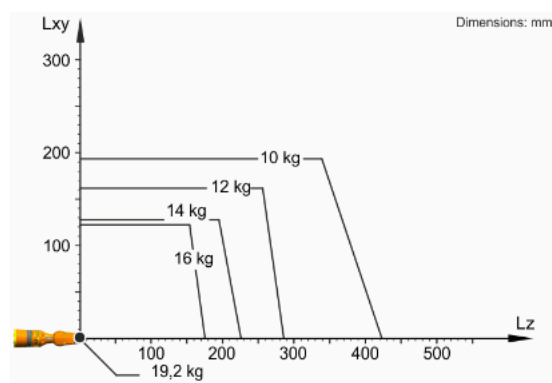
Teach pendant

Teach pendant	KUKA smartPAD
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Workspace graphic

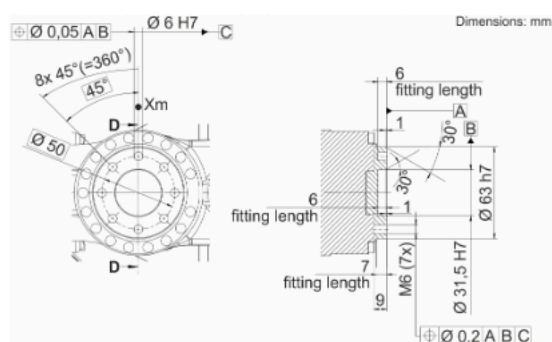


Payload diagram



The KR 16 R2010 is designed for a rated payload of 16 kg in order to optimize the dynamic performance of the robot. With reduced load center distances, higher loads up to the maximum payload may be used. The specific load case must be verified using KUKA.Load. For further consultation, please contact KUKA Support.

Mounting flange



IRB 1600

The highest performance 10 kg robot



Performance is often a trade off, optimizing for speed or accuracy. With ABB's IRB 1600, you don't have to choose. The robot's cycle times are shorter, sometimes half that of other robots, allowing you to increase throughput. Meanwhile, you will enjoy the work piece quality that only an ABB robot can offer. Extra everything.

Double your throughput

The IRB 1600 has up to 50 percent shorter cycle times than competing robots in material handling, machine tending and process applications. It speeds up and slows down faster than other robots, saving time while moving between tasks. This is possible due to ABB's patented second generation QuickMove motion control, combined with the robot's strong motors and low friction losses in the spur gears.

No more cutting corners

At high speed, most robots will cut corners. With the IRB 1600, the path will be the same regardless of speed, thanks to the robot's unique combination of brains and brawn. Intelligent second generation TrueMove motion control means that "what-you-program-is-what-you-get". Add muscle – a heavy and stiff design – low vibrations and low friction – and you have a robot that will deliver consistently high work piece quality, high yield and few rejects.

Outstanding reliability

The IRB 1600 offers outstanding reliability, even in the toughest environments and the most demanding 24/7 duty cycles. The entire manipulator is IP 54 classed and sensitive parts are IP 67 classed as standard.

The optional protection Foundry Plus offers IP 67, special paint, rust protection and is tailor made for tough foundry environments. The rigid and heavy design combined with spur gears, make the robot extremely robust. Smart collision detection software further adds to the robot's outstanding reliability.

Easy to integrate

Mounting is fully flexible: on a shelf, on the wall, tilted or inverted. By choosing the compact short-arm version with the 1.2 m reach, you can even fit the IRB 1600 inside a machine, while ensuring sufficient payload as the maximum total load is as high as 36 kg. Sustainable and healthy Low friction spur gears, and no unnecessary moves due to QuickMove and TrueMove, reduces power consumption down to 0.58 kW at max speed, and even less at low speeds. The airborne noise level of just <70 dB (A) secures a healthy sound environment.

Main applications

- Assembly
- Arc Welding
- Material Handling
- Machine Tending
- Material Removal
- Cleaning/Spraying
- Dispensing
- Packing

Specification

Robot version	Reach (m)	Payload (kg)	Armload (kg)
IRB 1600-6/1.2	1.2	6	30.5
IRB 1600-6/1.45	1.45	6	30.5
IRB 1600-10/1.2	1.2	10	20.5
IRB 1600-10/1.45	1.45	10	20.5
Number of axes	6+3 external (up to 36 with MultiMove)		
Protection	Standard IP54 Option FoundryPlus 2 (IP67)		
Mounting	Floor, wall, shelf, tilted, inverted		
Controller	IRC5 Single Cabinet, Compact, OmniCore C30, C90XT, V250XT, V400XT		

Performance (according to ISO 9283)

	Position repeatability	Path repeatability
IRB 1600-6/1.2	0.02 mm	0.13 mm
IRB 1600-6/1.45	0.02 mm	0.19 mm
IRB 1600-10/1.2	0.02 mm	0.06 mm
IRB 1600-10/1.45	0.05 mm	0.13 mm

Technical information

Electrical Connections

Supply voltage	200-600 V, 50-60 Hz
Energy consumption	0.58 kW

Physical

Robot base	484 x 648 mm
Robot height	
IRB 1600-6/1.2	1069 mm
IRB 1600-10/1.2	1069 mm
IRB 1600-6/1.45	1294 mm
IRB 1600-10/1.45	1294 mm
Robot weight	250 kg

Environment

Ambient temperature for mechanical unit	
During operation	+ 5°C (41°F) to + 45°C (113°F)
During transportation and storage	- 25°C (- 13°F) to + 55°C (131°F)
During short periods (max. 24 h)	up to + 70°C (158°F)
Relative humidity	Max. 95%
Safety	Double circuits with supervisions, emergency stops and safety functions, 3-position enable device
Emission	EMC/EMI shielded

Data and dimensions may be changed without notice.

Movement

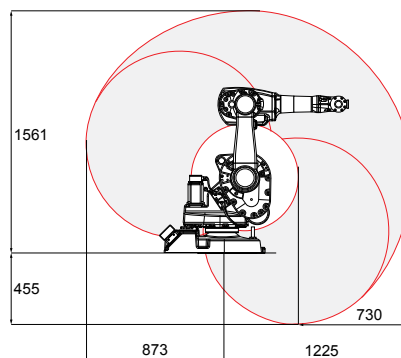
Axis movement	Working range IRB 1600-6/1.2, IRB 1600-10/1.2	Working range IRB 1600-6/1.45, IRB 1600-10/1.45
Axis 1 rotation	+180° to -180°	+180° to -180°
Axis 2 arm	+110° to -63° +136° to -63° ¹	+120° to -90° +150° to -90° ²
Axis 3 arm	+55° to -235°	+65° to -245°
Axis 4 rotation	Default: +200° to -200° Max. rev: +190° to -190°	Default: +200° to -200° Max. rev: +190° to -190°
Axis 5 bend	+115° to -115°	+115° to -115°
Axis 6 turn	Default: +400° to -400° Max. rev: +288 to -288	Default: +400° to -400° Max. rev: +288 to -288

¹With axis 1 limited to ±100° ²With axis 1 limited to ±95°

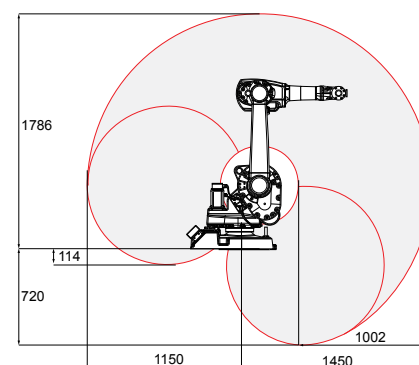
Axis max. speed

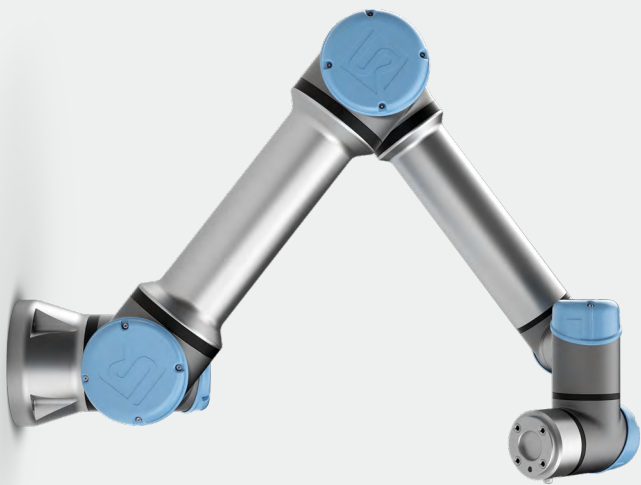
	IRB 1600-6/1.2, IRB 1600-6/1.45	IRB 1600-10/1.2, IRB 1600-10/1.45
Axis 1 rotation	150°/s	180°/s
Axis 2 arm	160°/s	180°/s
Axis 3 arm	170°/s	185°/s
Axis 4 rotation	320°/s	385°/s
Axis 5 bend	400°/s	400°/s
Axis 6 turn	460°/s	460°/s

Working range, IRB 1600-6/1.2, IRB 1600-10/1.2



Working range, IRB 1600-6/1.45, IRB 1600-10/1.45





UR5e

Technical Specification

The UR5e brings ultimate flexibility to medium-duty applications with a payload of up to 5 kg and a reach of 850 mm.

Today, more than 90,000 UR collaborative industrial robots have been delivered to customers across industries and around the world. The e-Series of cobots brings incredible flexibility and unparalleled ease of use to your application.

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UR5e

Specification

Payload	5 kg (11 lbs)
Reach	850 mm (33.5 in)
Degrees of freedom	6 rotating joints
Programming	12 inch touchscreen with PolyScope graphical user interface
Power consumption (average)	570 W
Maximum power	200 W
Moderate operating settings	Ambient temperature: 0-50°C (32-122°F)
Operating temperature range	17 configurable safety functions
Safety functions	EN ISO 13849-1 (PLd Category 3) and EN ISO 10218-1

Performance

Force sensing, tool flange/torque sensor	Force, x-y-z	Torque, x-y-z
Range	± 50.0 N	± 10.0 Nm
Precision	± 3.5 N	± 0.2 Nm
Accuracy	± 4.0 N	± 0.3 Nm

Movement

Max TCP speed	4 m/s	
Pose Repeatability per ISO 9283	± 0.03 mm	
Axis movement	Working range	Maximum speed
Base	± 360°	± 180°/s
Shoulder	± 360°	± 180°/s
Elbow	± 360°	± 180°/s
Wrist 1	± 360°	± 180°/s
Wrist 2	± 360°	± 180°/s
Wrist 3	± 360°	± 180°/s

Teach Pendant

Features

IP classification	IP54
Humidity	≤ 90% RH (non-condensing)
Display resolution	1280 x 800 pixels

Physical

Materials	Plastic
Teach Pendant size (W x H x D)	300 mm x 231 mm x 50 mm (11.8 in x 9.1 in x 1.97 in)
Weight	1.8 kg (3.961 lbs) including 1 m of teach pendant cable
Cable length	4.5 m (177.17 in)

Control Box

Features

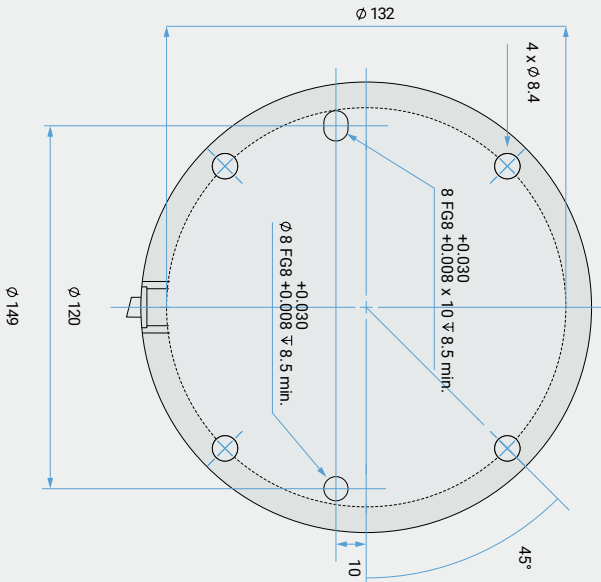
IP Classification	IP44
Operating temperature range*	Ambient temperature: 0-50°C (32-122°F)
Humidity	≤ 90% RH (non-condensing)
I/O Ports	Digital In 16 Digital Out 16 Analog In 2 Analog Out 2 Quadrature Digital Inputs 4
I/O Power Supply	24V, 2A
Industrial Protocols	Modbus-TCP (Client/Server) Ethernet/IP Adapter PROFINET device/PROFIsafe ROS/ROS2
Hardware Interfaces	Ethernet 1 Gb/s USB 2.0, USB 3.0 Mini DisplayPort Injection Moulding Machine Interface (SPI AN-146 & Euromap-67)
Power Source	100-240 VAC, 47-440 Hz

Physical

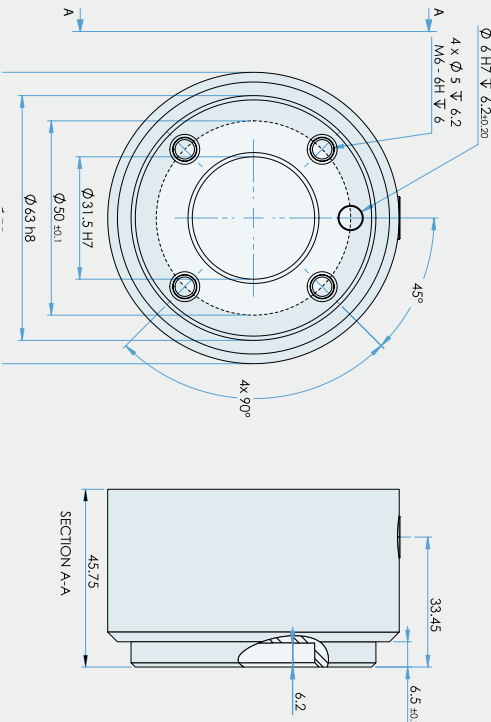
Control Box Size (W x H x D)	460 mm x 449 mm x 254 mm (18.2 in x 17.6 in x 10 in)
Weight	12 kg (26.5 lbs)
Materials	Powder coated steel

The control box is also available in an OEM version

Interface for UR5e base flange



Tool flange for UR5e



UR5e payload curve

